

Read a baby age in no of days and find the baby age in years, months, weeks and days.

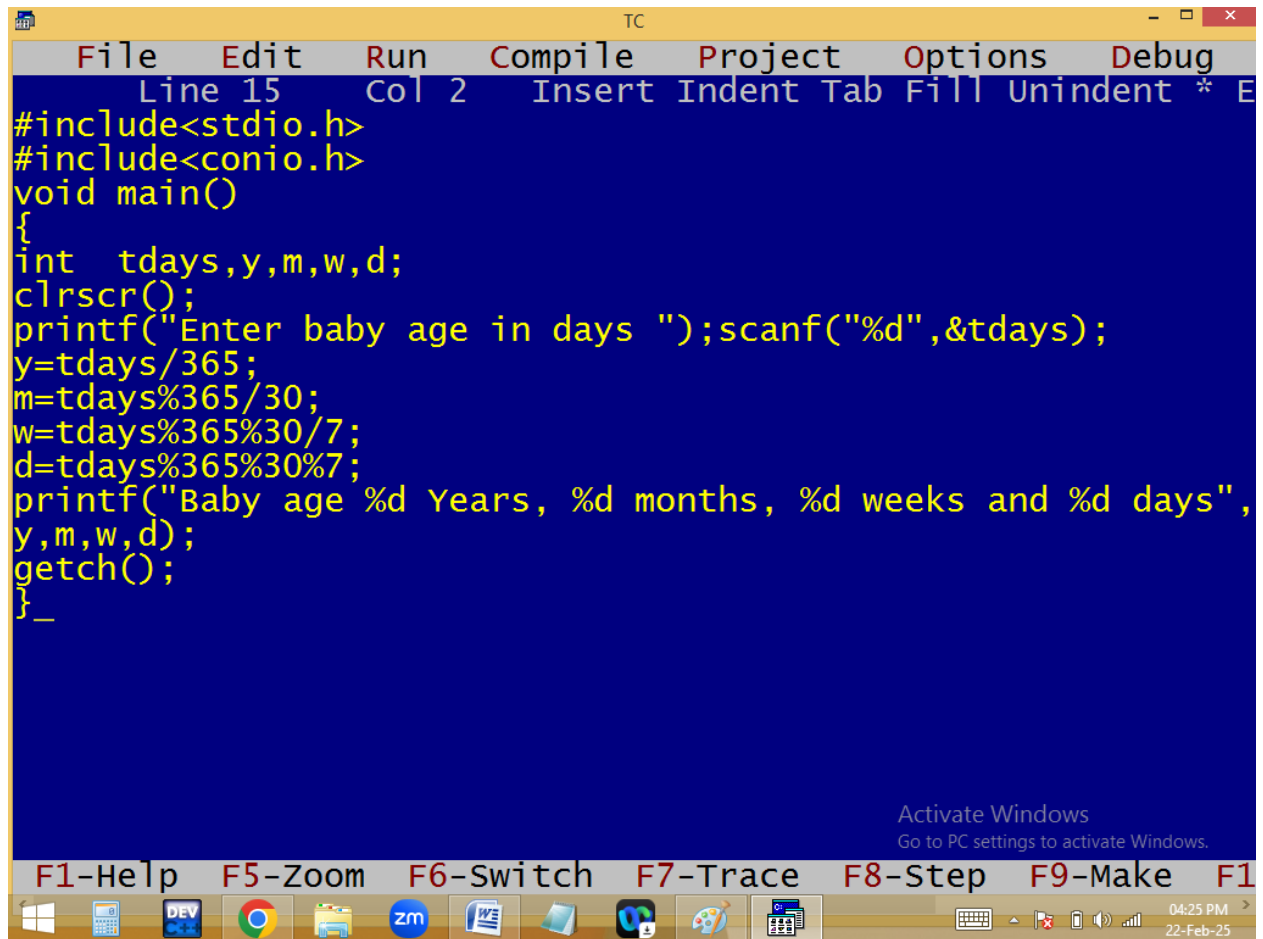
$$y = 500/365=1$$

$$m = 500\%365=135/30=4$$

$$w = 500\%365=135\%30=15/7=2$$

$$d = 500\%365=135\%30=15\%7=1$$

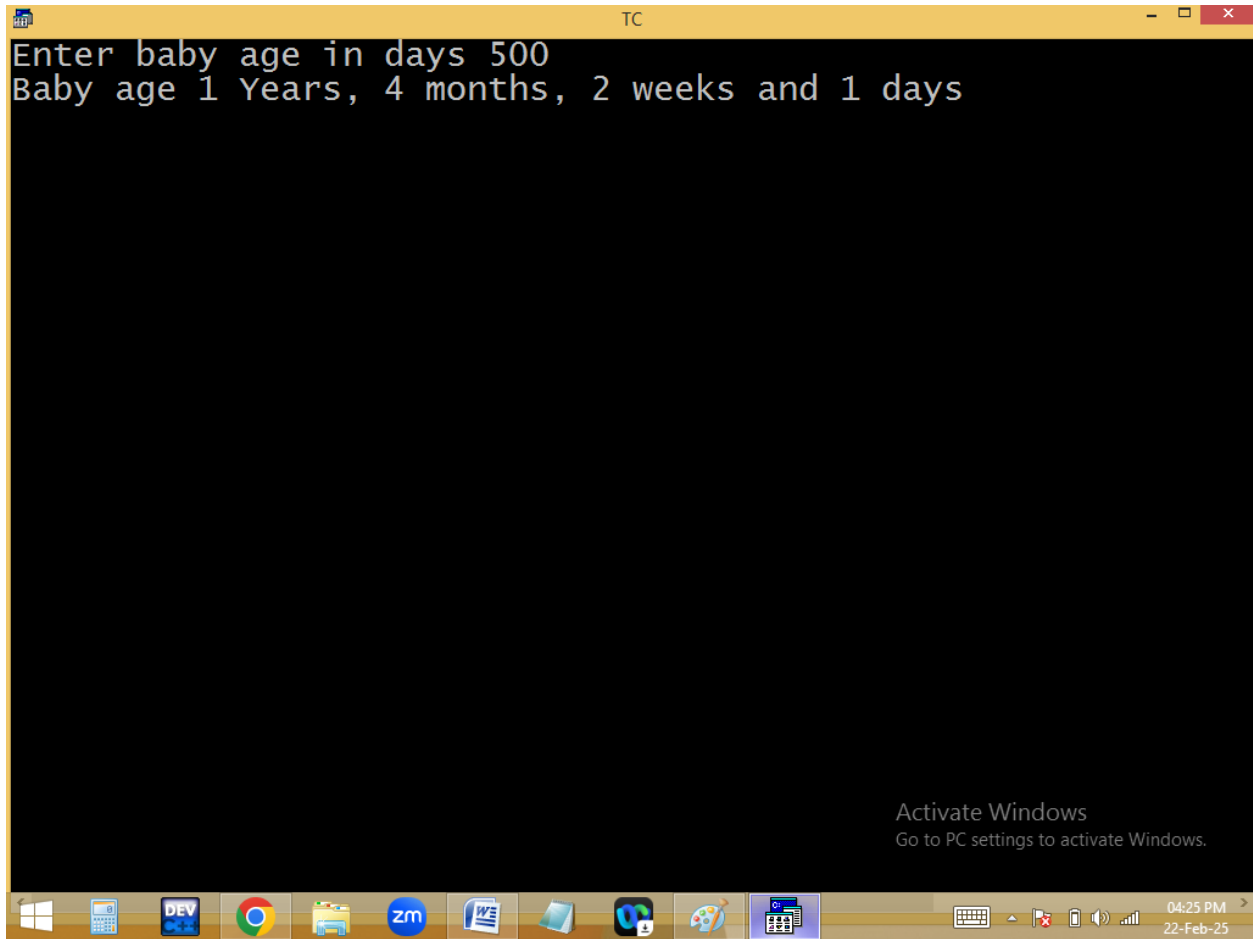
$$\begin{array}{r} \text{tdays} \\ \hline 365 \overline{) 500} \quad (1-y \\ \underline{365} \\ 135 \\ 30 \overline{) 135} \quad (4-m \\ \underline{120} \\ 15 \\ 7 \overline{) 15} \quad (2-w \\ \underline{14} \\ 1-d \end{array}$$

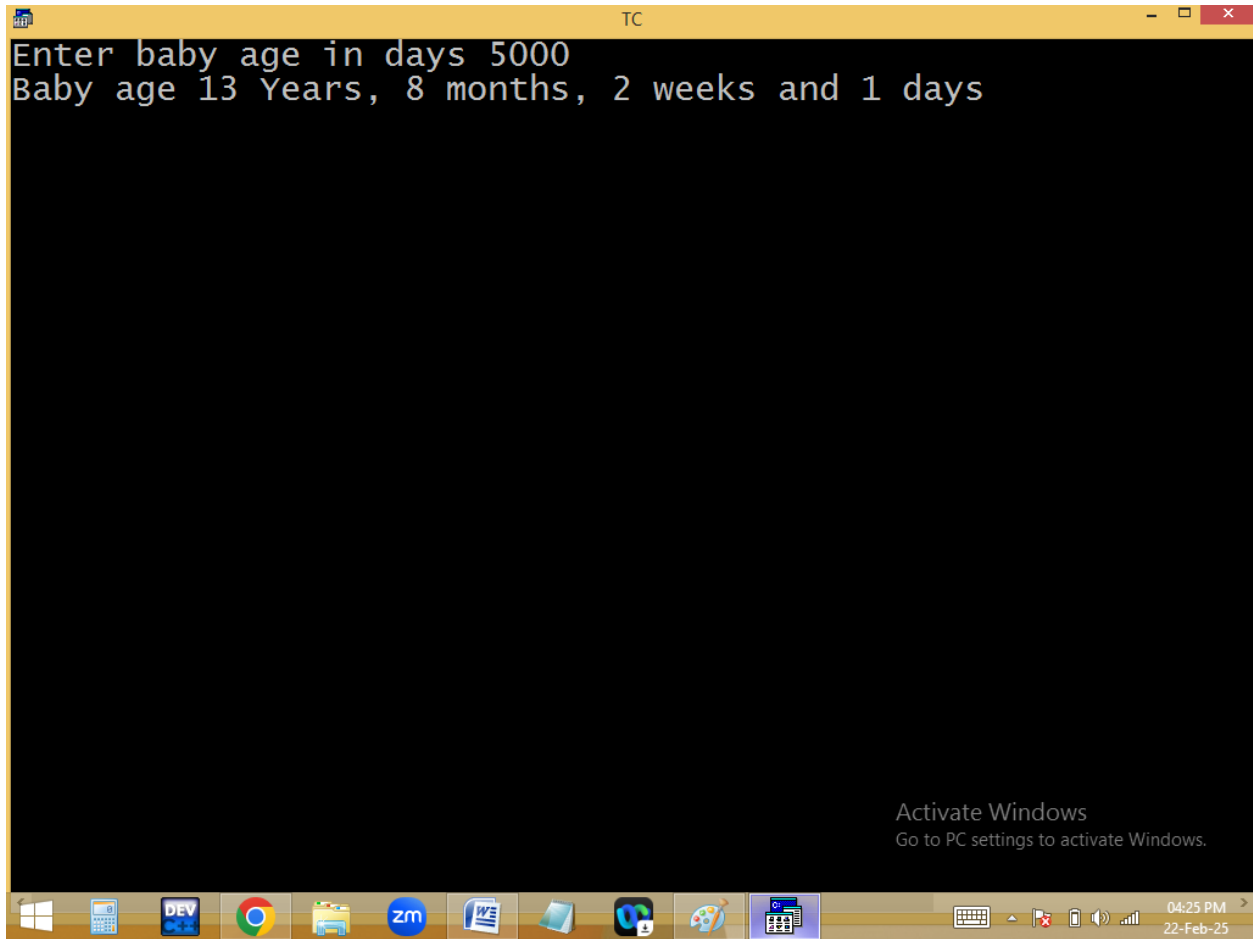


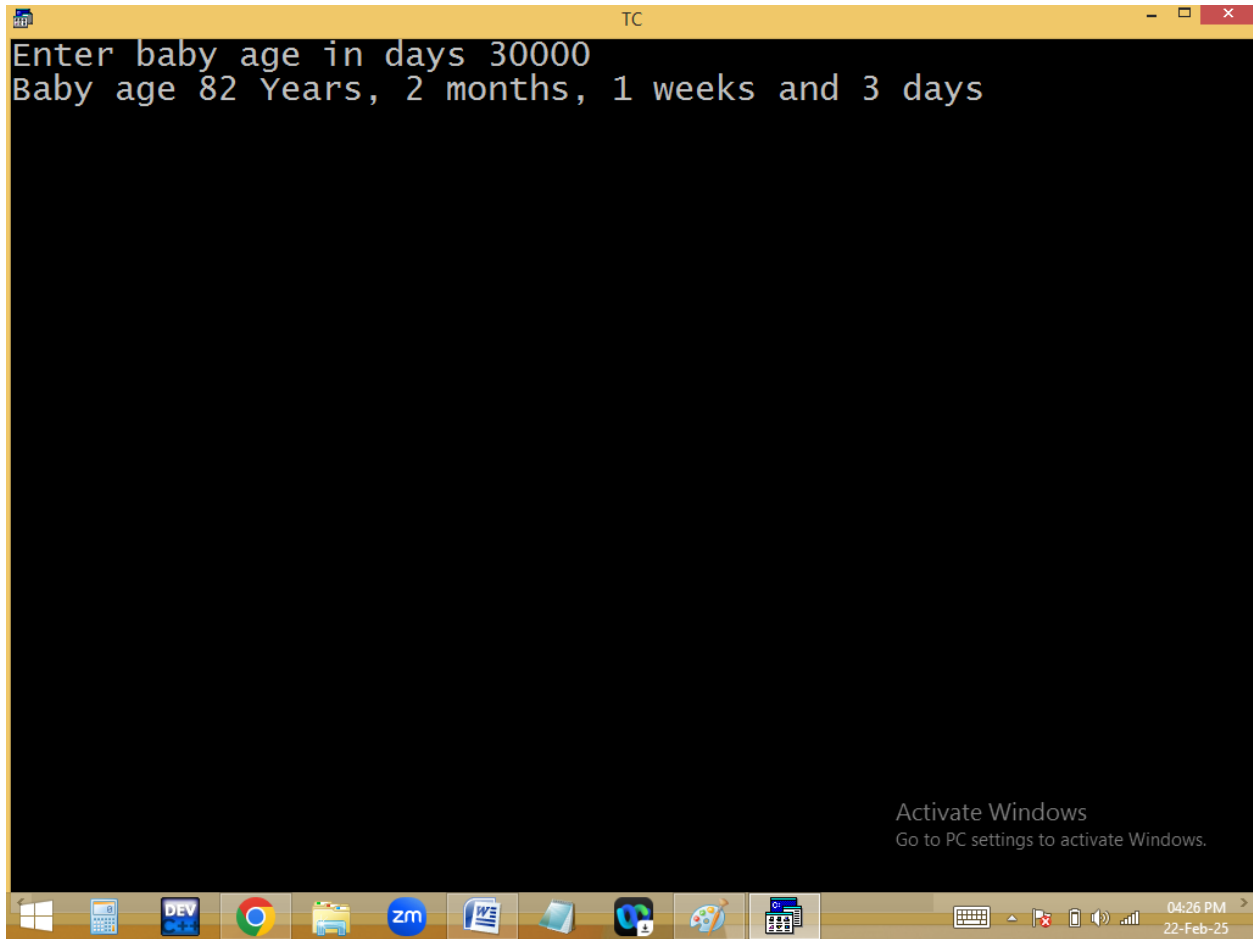
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 15", "Col 2", and "Insert Indent Tab Fill Unindent * E". The main editing area has a blue background with yellow text containing the following C code:

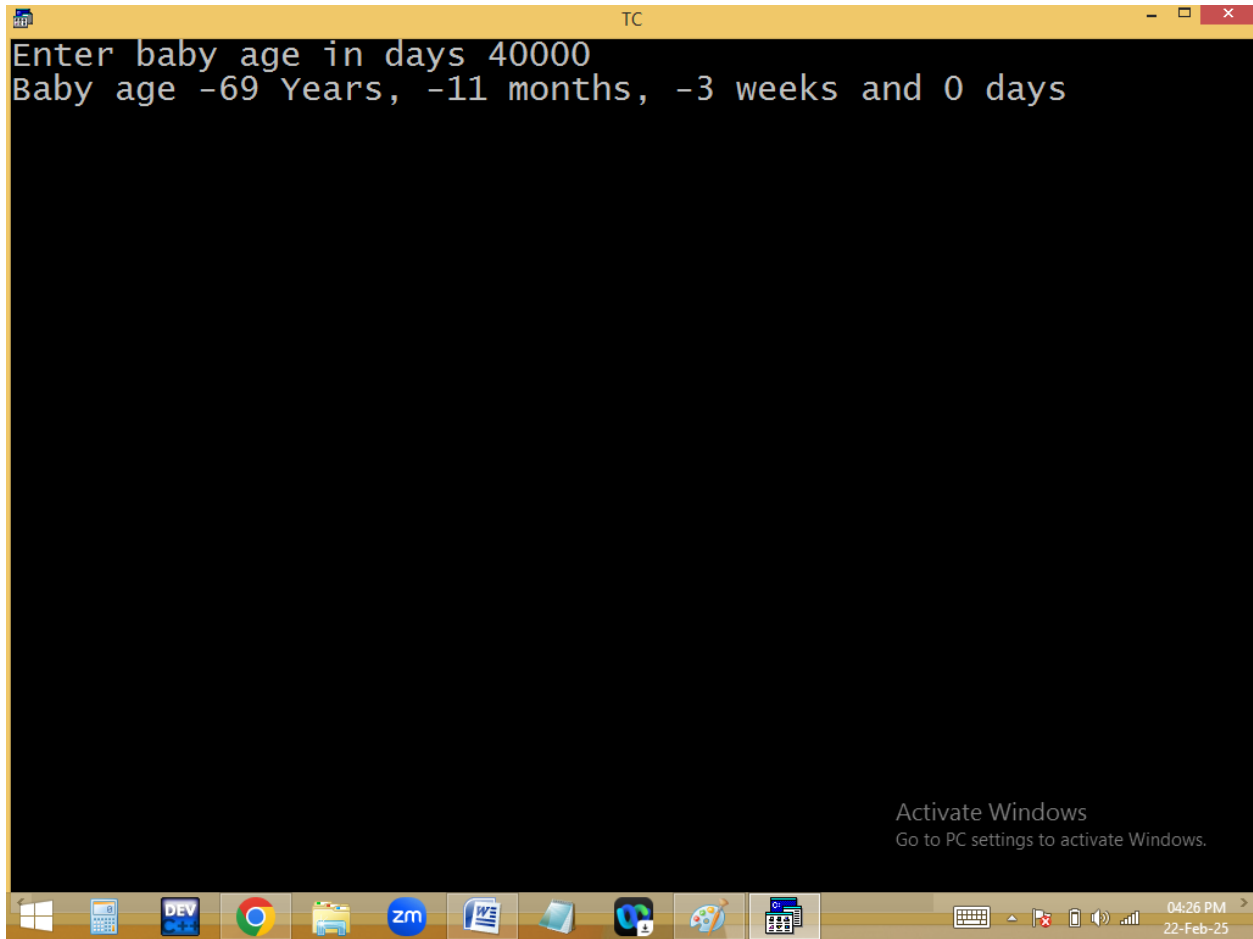
```
#include<stdio.h>
#include<conio.h>
void main()
{
int  tdays,y,m,w,d;
clrscr();
printf("Enter baby age in days ");scanf("%d",&tdays);
y=tdays/365;
m=tdays%365/30;
w=tdays%365%30/7;
d=tdays%365%30%7;
printf("Baby age %d Years, %d months, %d weeks and %d days",
y,m,w,d);
getch();
}_
```

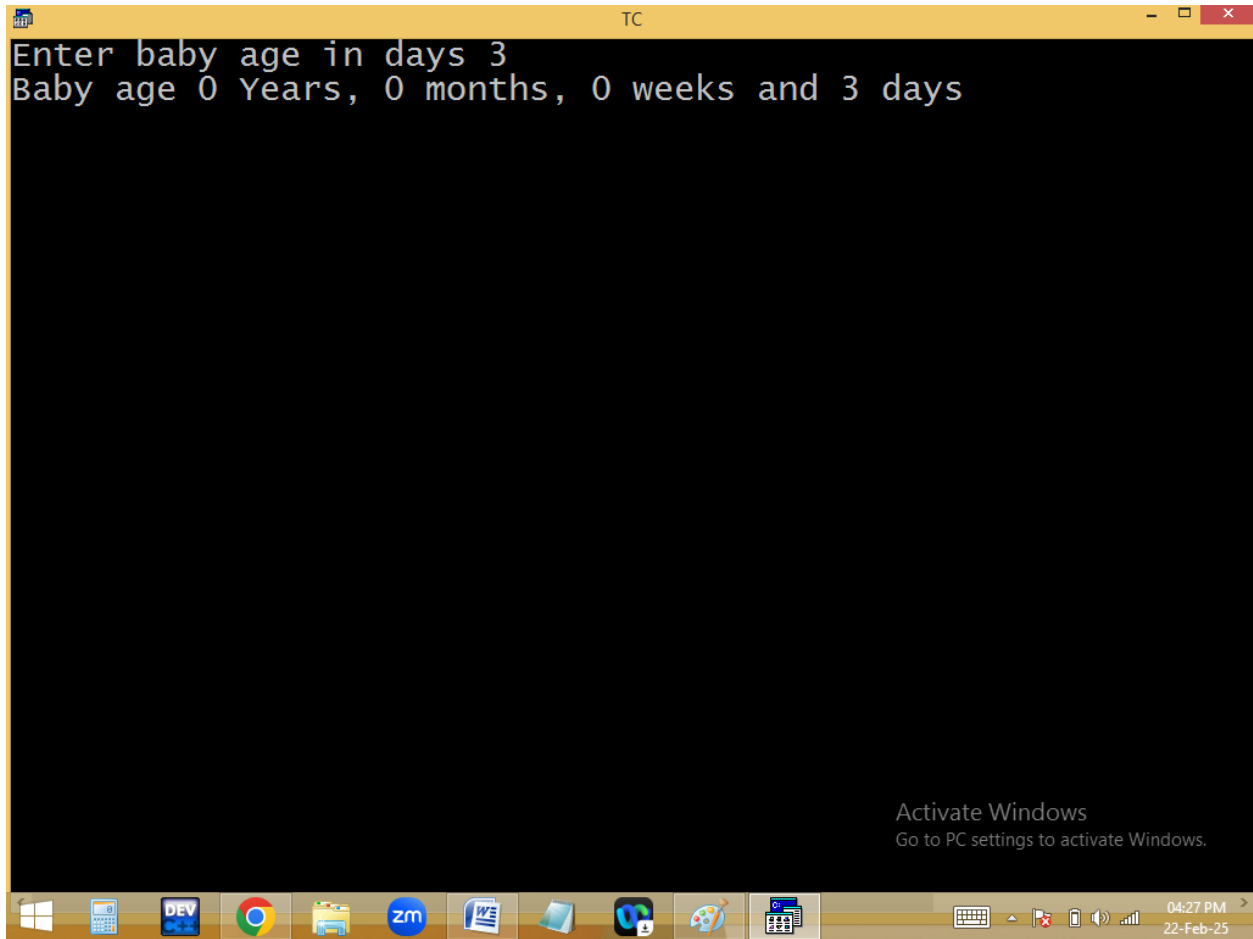
At the bottom of the IDE window, there is a row of function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". Below this, the Windows taskbar is visible with icons for the Start menu, Task View, File Explorer, Microsoft Edge, Zoom, Word, and several other applications. The system tray on the right shows the time as "04:25 PM" and the date as "22-Feb-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

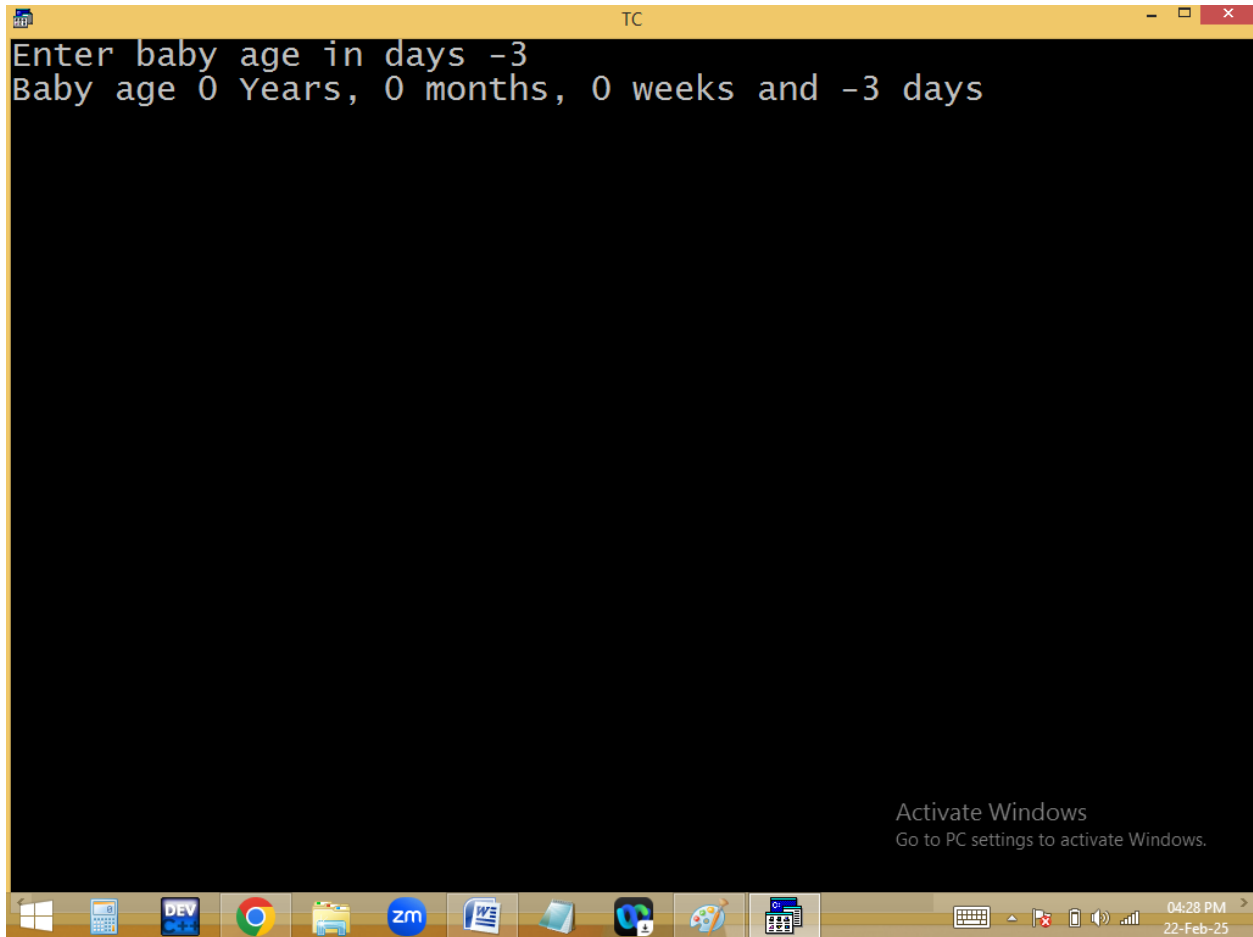










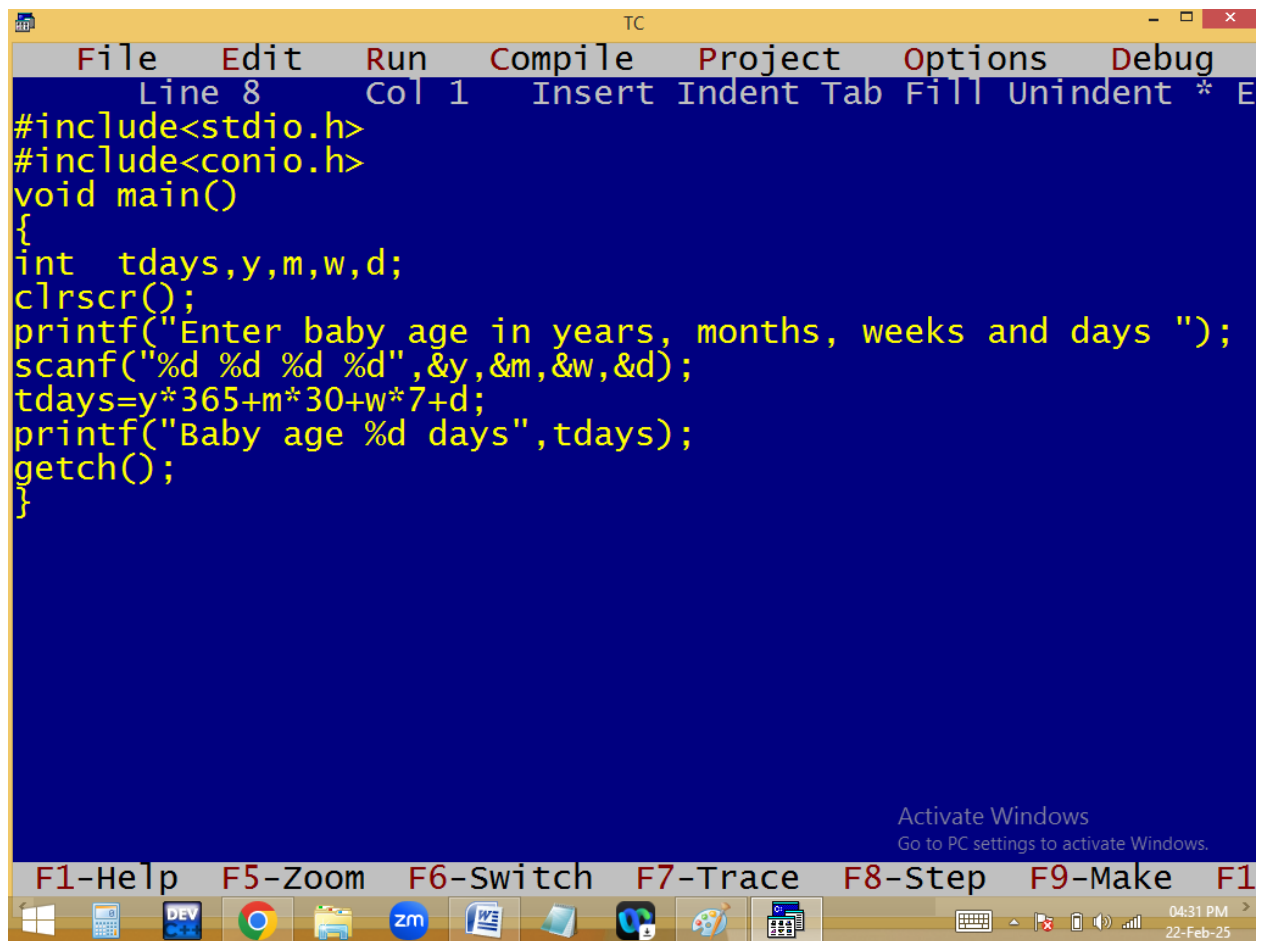


**Read a baby age in years, months, weeks and days.
Find the baby age in days.**

1 year 4 months 2 weeks and 1 day.

Baby age 500 days

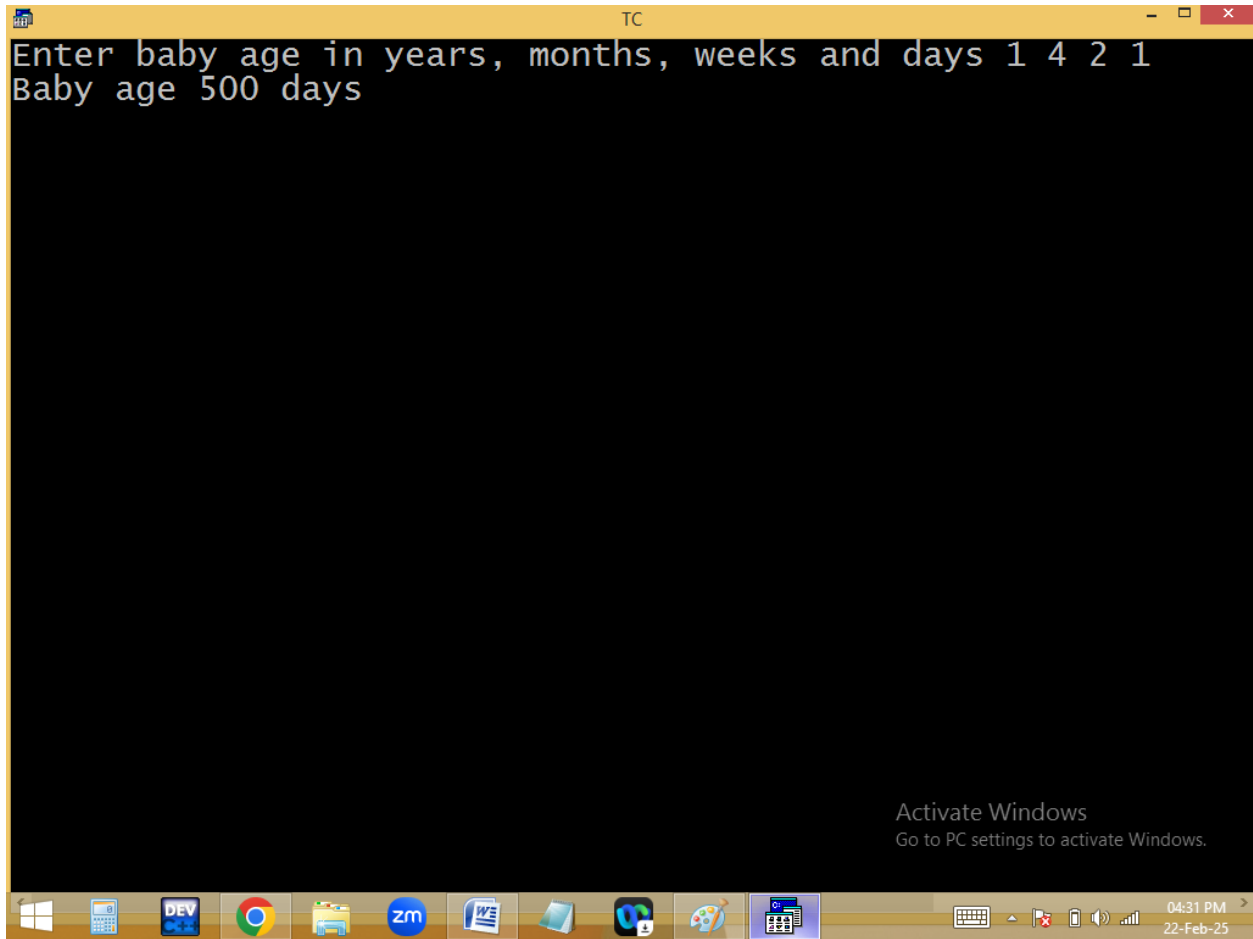
$$1 * 365 + 4 * 30 + 2 * 7 + 1 = 500 \text{ days}$$

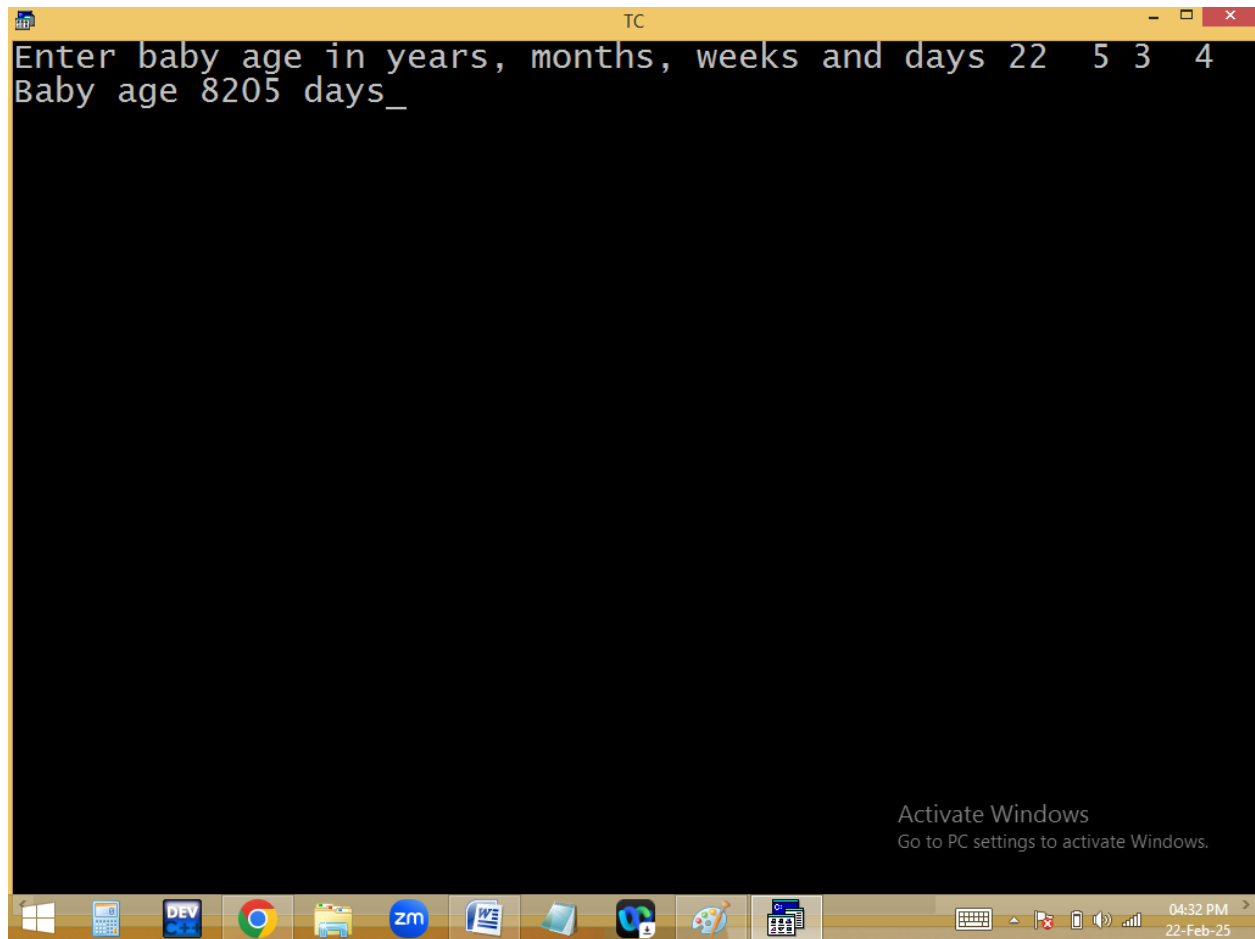


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 8 Col 1" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, and *. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int  tdays,y,m,w,d;
clrscr();
printf("Enter baby age in years, months, weeks and days ");
scanf("%d %d %d %d",&y,&m,&w,&d);
tdays=y*365+m*30+w*7+d;
printf("Baby age %d days",tdays);
getch();
}
```

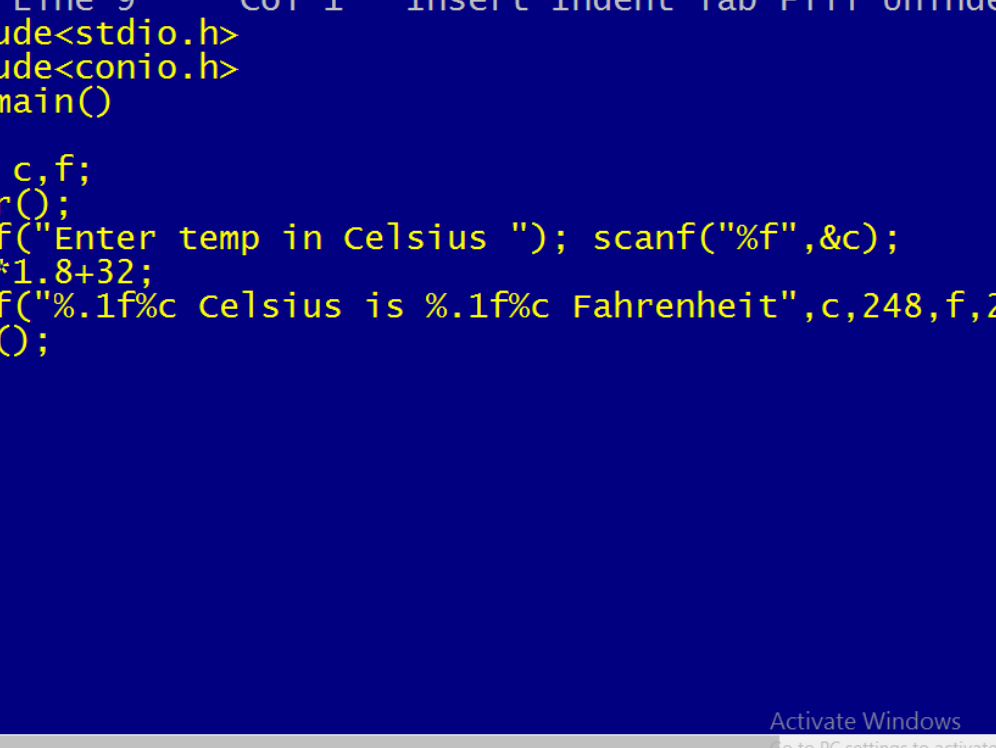
At the bottom of the IDE window, there is a row of function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. Below this is the Windows taskbar with icons for Windows, Calculator, DEV C++, Google Chrome, File Explorer, Zoom, Notepad, and several other applications. The system tray on the right shows the time as 04:31 PM and the date as 22-Feb-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.





Celsius to Fahrenheit conversion:

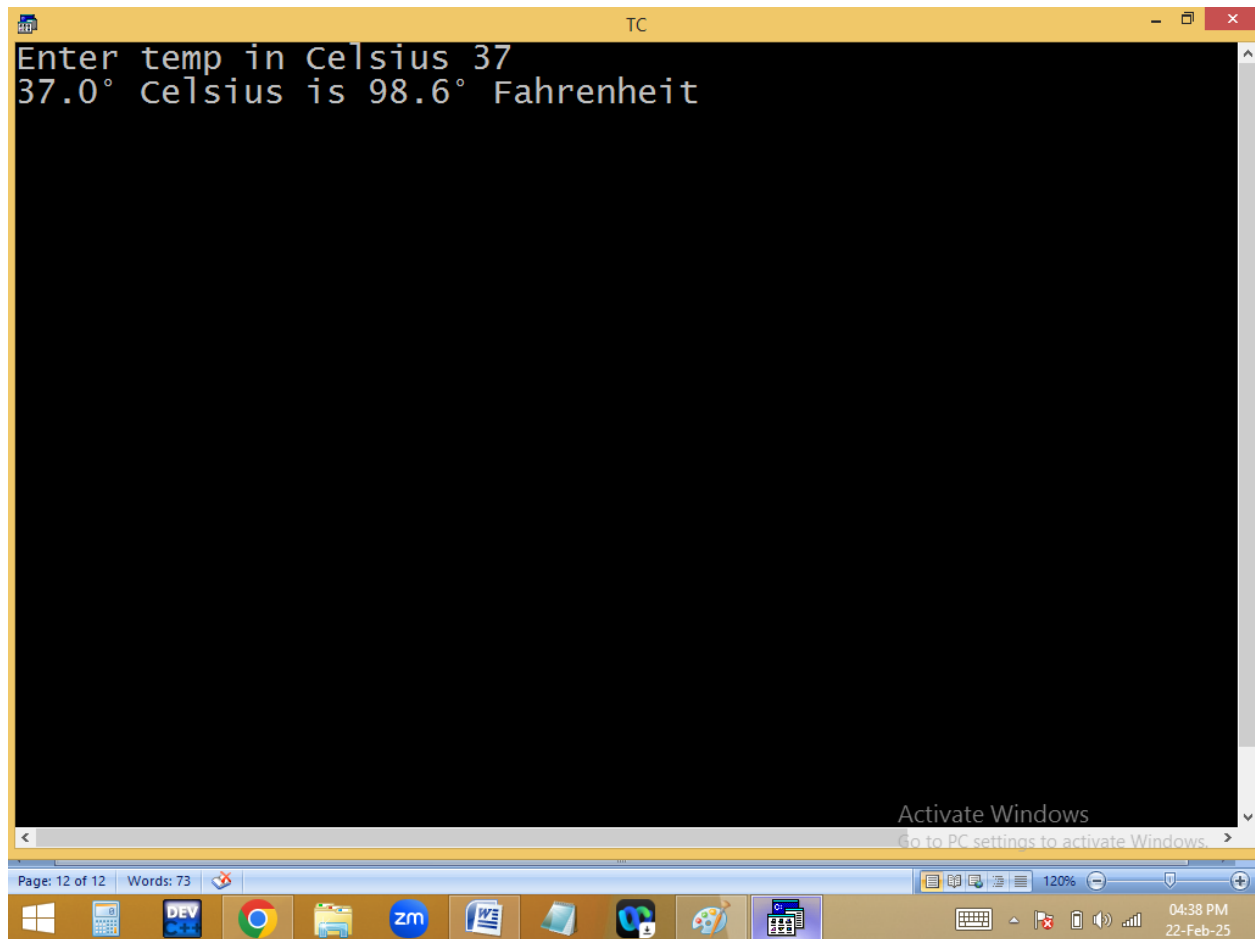
$$f = c * 1.8 + 32$$



The screenshot shows the Turbo C++ (TC) IDE interface. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar indicates "Line 9", "Col 1", and a list of actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main editing area has a dark blue background with yellow text. The code is as follows:

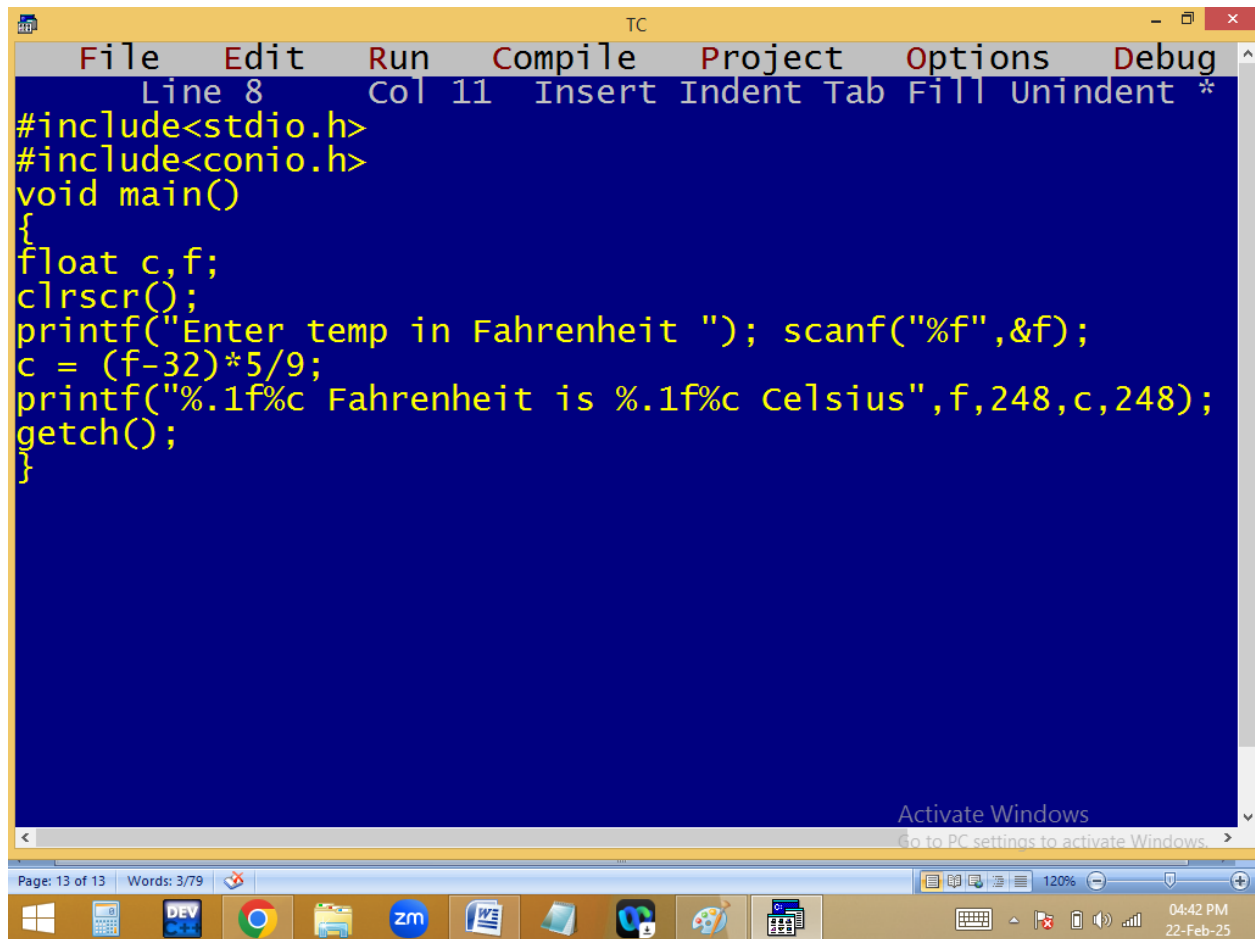
```
#include<stdio.h>
#include<conio.h>
void main()
{
float c,f;
clrscr();
printf("Enter temp in Celsius "); scanf("%f",&c);
f = c*1.8+32;
printf("%.1f%c Celsius is %.1f%c Fahrenheit",c,248,f,248);
getch();
}
```

At the bottom of the IDE window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows". The Windows taskbar is visible at the very bottom, showing the Start button, several application icons (including DEV, Chrome, and Zoom), and the system clock displaying "04:38 PM" and "22-Feb-25".



Fahrenheit to Celsius:

$$C = f - 32 * 5/9$$



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 8", "Col 11", and lists editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main editing area has a dark blue background with yellow text. It contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
float c,f;
clrscr();
printf("Enter temp in Fahrenheit "); scanf("%f",&f);
c = (f-32)*5/9;
printf("%.1f%c Fahrenheit is %.1f%c Celsius",f,248,c,248);
getch();
}
```

At the bottom of the IDE window, there is a status bar showing "Page: 13 of 13" and "Words: 3/79". Below the IDE window is the Windows taskbar, which includes icons for the Start button, Task View, File Explorer, Google Chrome, Zoom, Word, and other applications. The system tray on the right shows the date and time as "04:42 PM 22-Feb-25".

```
TC
Enter temp in Fahrenheit 98.6
98.6° Fahrenheit is 37.0° Celsius
```

Activate Windows
Go to PC settings to activate Windows.

The screenshot shows a Windows 10 desktop. A Turbo C++ (TC) window is open, displaying a program that prompts the user to enter a temperature in Fahrenheit. The user has entered '102.3', and the program has calculated and displayed '102.3° Fahrenheit is 39.1° Celsius'. Below the TC window, a Microsoft Word document is open, showing 'Page: 15 of 15' and 'Words: 79'. The taskbar at the bottom contains icons for various applications including the Start menu, File Explorer, Zoom, Word, and several utility icons. The system clock in the bottom right corner indicates the time is 04:42 PM on 22-Feb-25.

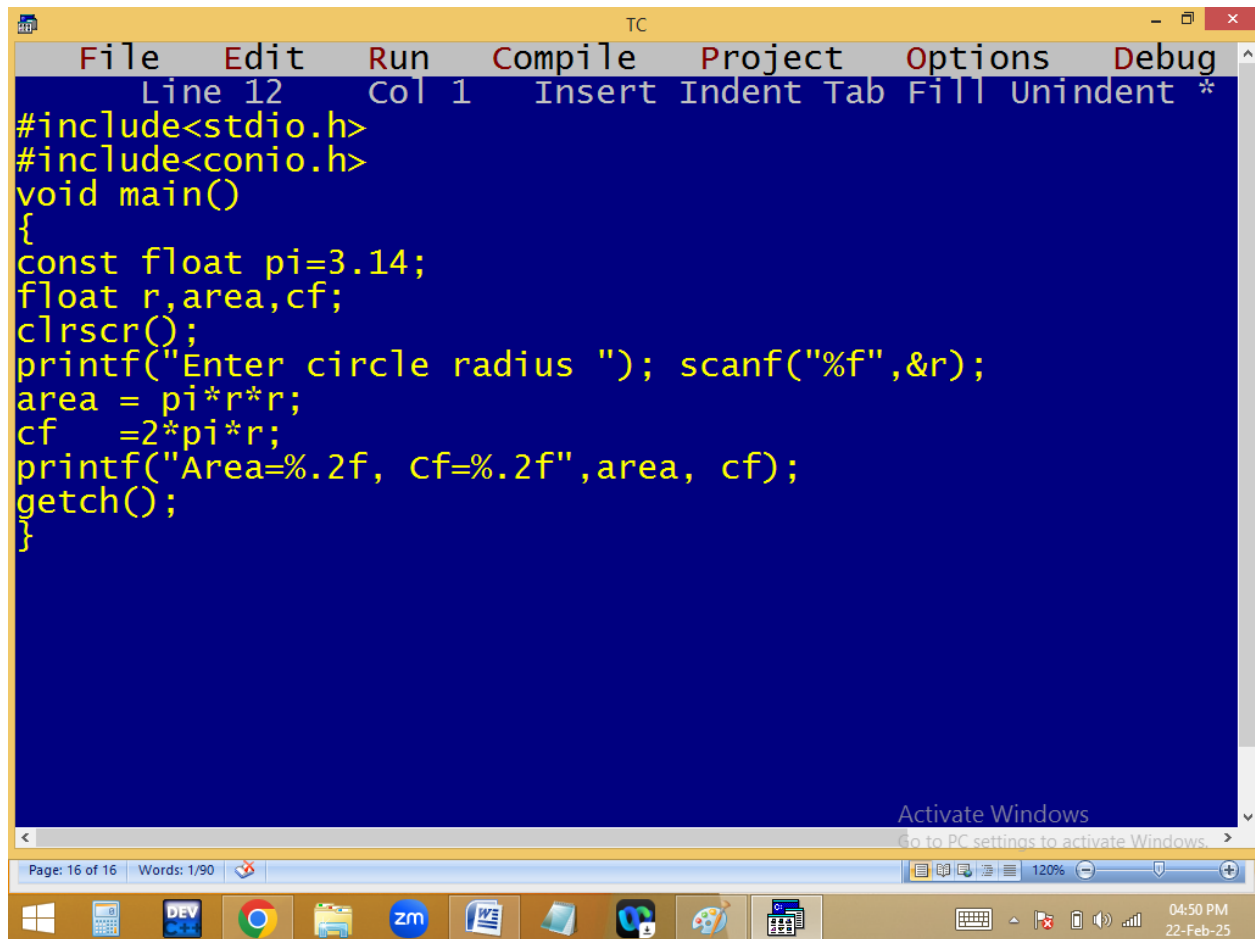
```
TC
Enter temp in Fahrenheit 102.3
102.3° Fahrenheit is 39.1° Celsius_

Activate Windows
Go to PC settings to activate Windows.
```

Page: 15 of 15 | Words: 79

04:42 PM
22-Feb-25

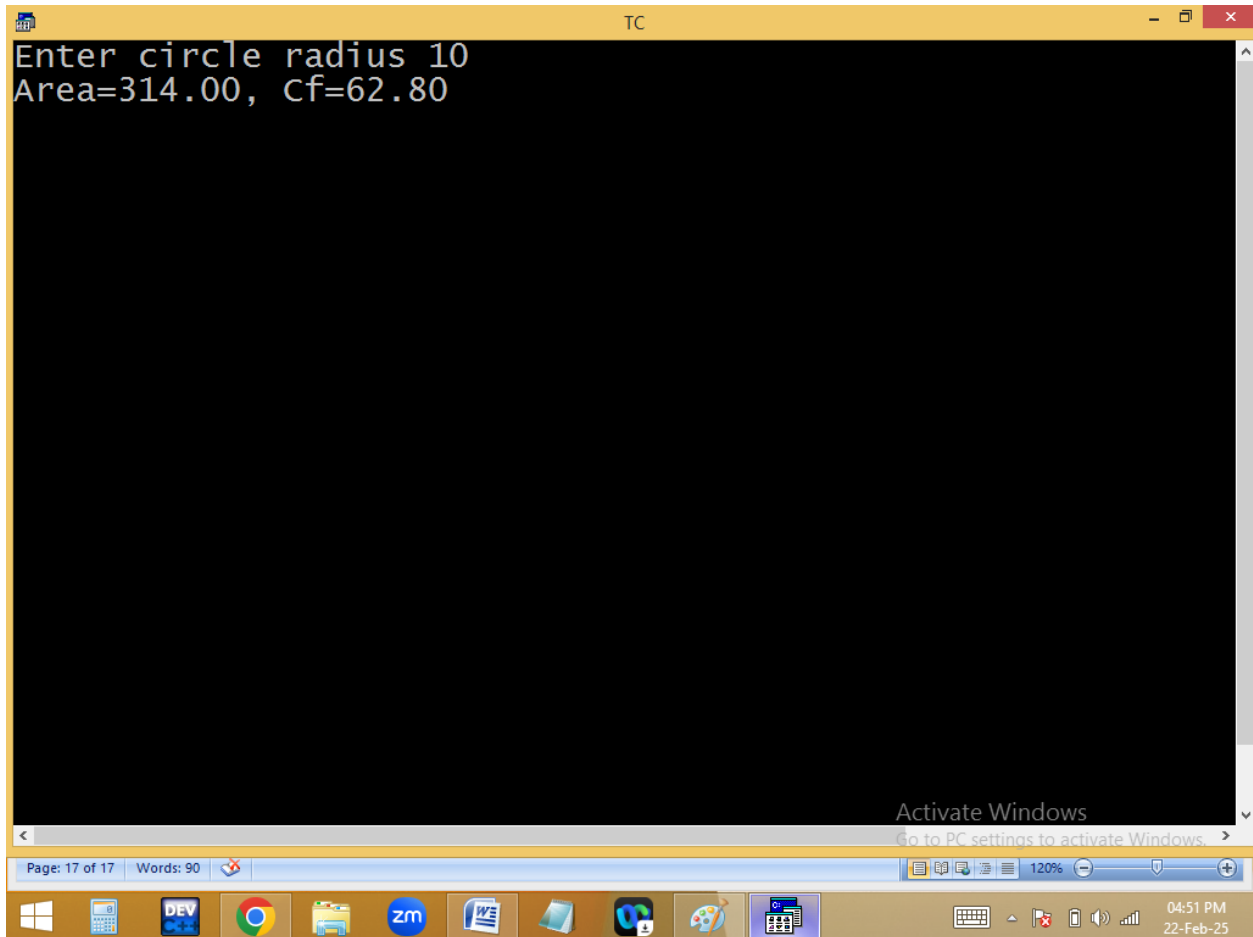
Read radius of a circle and find the area and circumference.

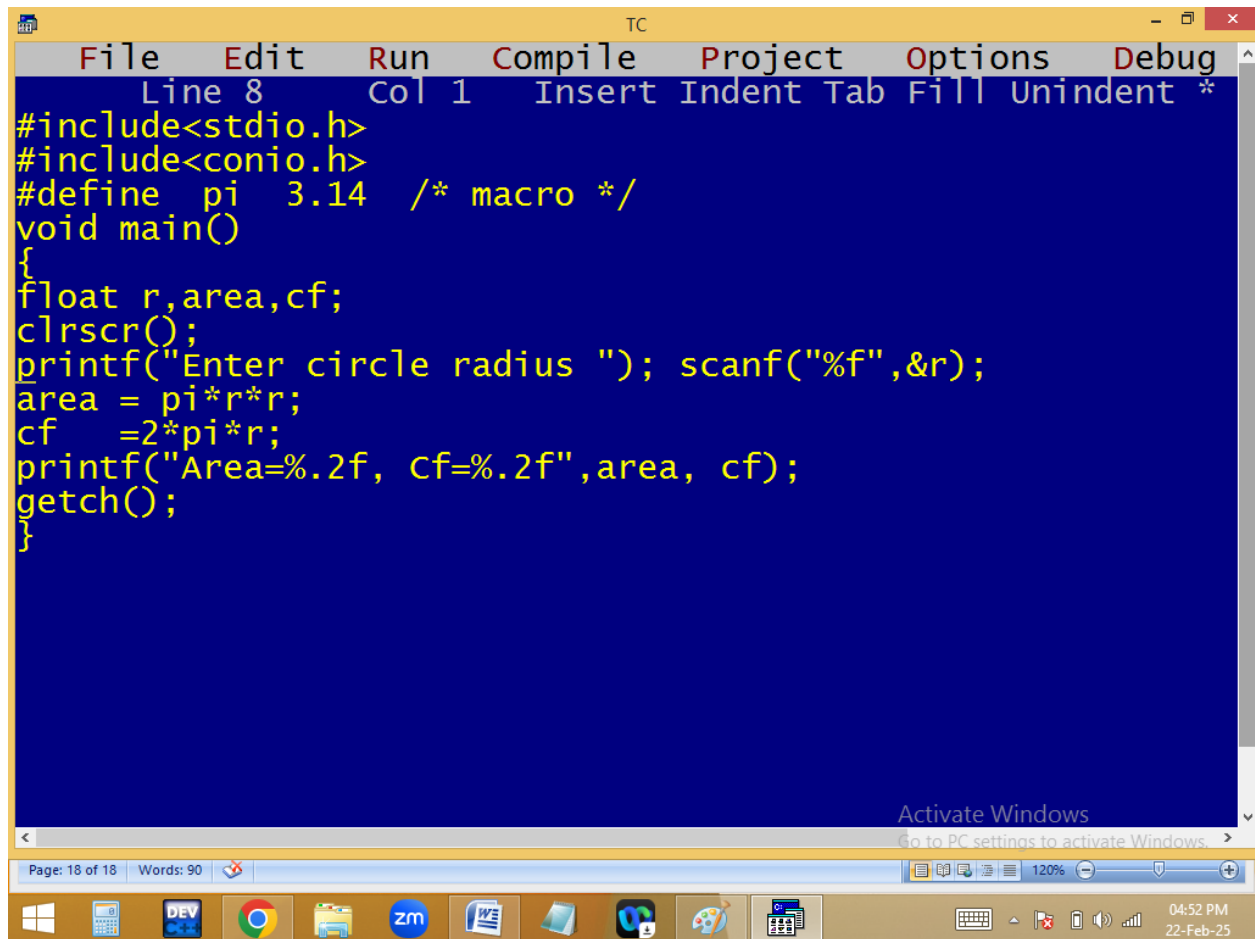


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard Windows window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 12", "Col 1", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    const float pi=3.14;
    float r,area,cf;
    clrscr();
    printf("Enter circle radius "); scanf("%f",&r);
    area = pi*r*r;
    cf    =2*pi*r;
    printf("Area=%.2f, Cf=%.2f",area, cf);
    getch();
}
```

At the bottom of the window, there is a status bar with "Page: 16 of 16" and "Words: 1/90". To the right of this, there is a "120%" zoom level indicator. The Windows taskbar is visible at the very bottom, showing various application icons and the system clock displaying "04:50 PM 22-Feb-25".





The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 8", "Col 1", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main editing area has a dark blue background with yellow text. It contains a C program that calculates the area and circumference of a circle. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
#define pi 3.14 /* macro */
void main()
{
float r,area,cf;
clrscr();
printf("Enter circle radius "); scanf("%f",&r);
area = pi*r*r;
cf = 2*pi*r;
printf("Area=%.2f, Cf=%.2f",area, cf);
getch();
}
```

At the bottom of the IDE window, there is a status bar showing "Page: 18 of 18" and "Words: 90". Below the IDE window is the Windows taskbar, which includes icons for the Start button, Task View, File Explorer, Google Chrome, Zoom, Word, and other applications. The system tray on the right shows the date and time as "04:52 PM 22-Feb-25".

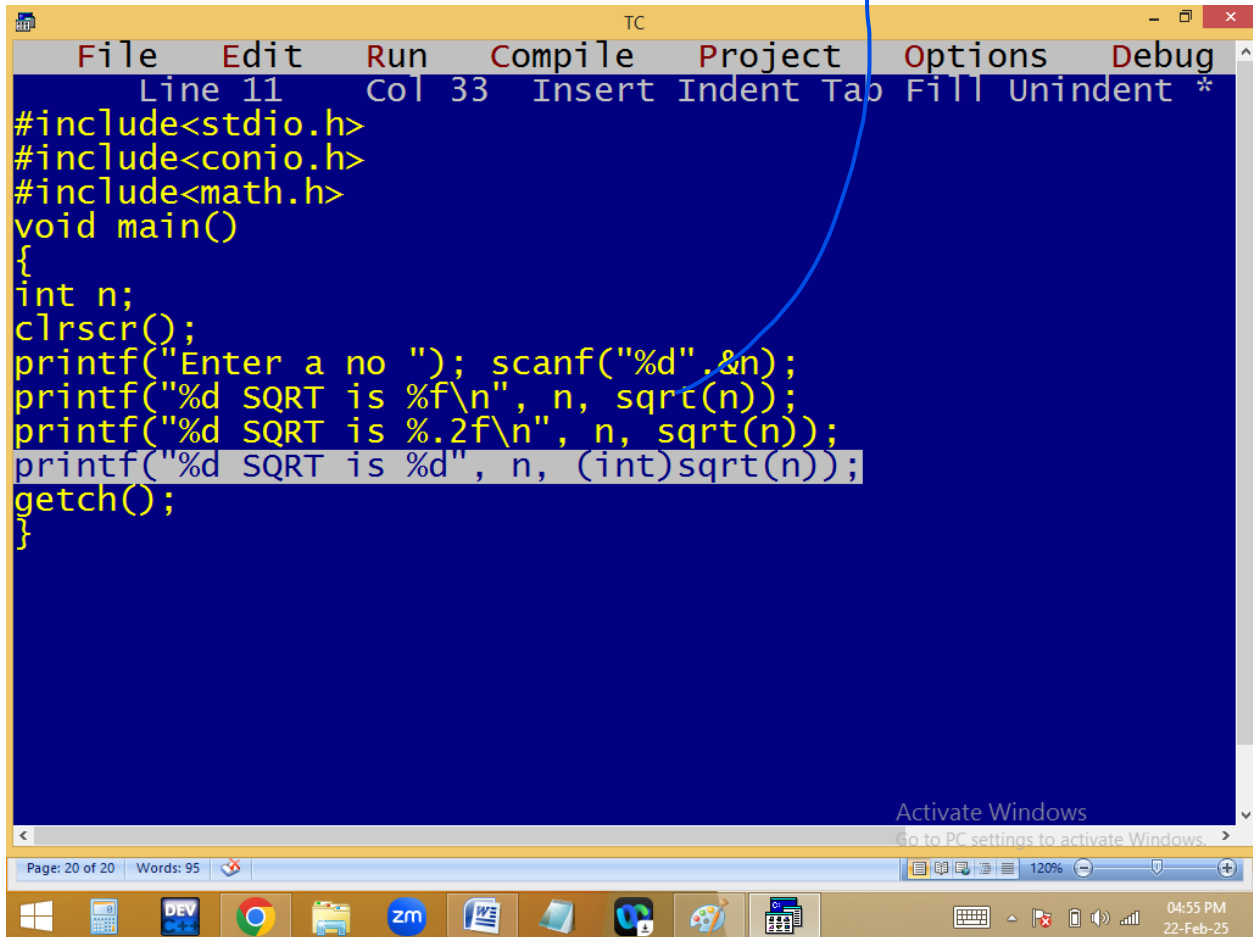
```
TC
Enter circle radius 2
Area=12.56, Cf=12.56
```

Page: 19 of 19 Words: 90 120%

04:52 PM 22-Feb-25

Finding SQRT of given no.

it returns float values



```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 33 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
int n;
clrscr();
printf("Enter a no "); scanf("%d",&n);
printf("%d SQRT is %f\n", n, sqrt(n));
printf("%d SQRT is %.2f\n", n, sqrt(n));
printf("%d SQRT is %d", n, (int)sqrt(n));
getch();
}
```

Page: 20 of 20 Words: 95 120% 04:55 PM 22-Feb-25

```
TC
Enter a no 100
100 SQRT is 10.000000
100 SQRT is 10.00
100 SQRT is 10
```

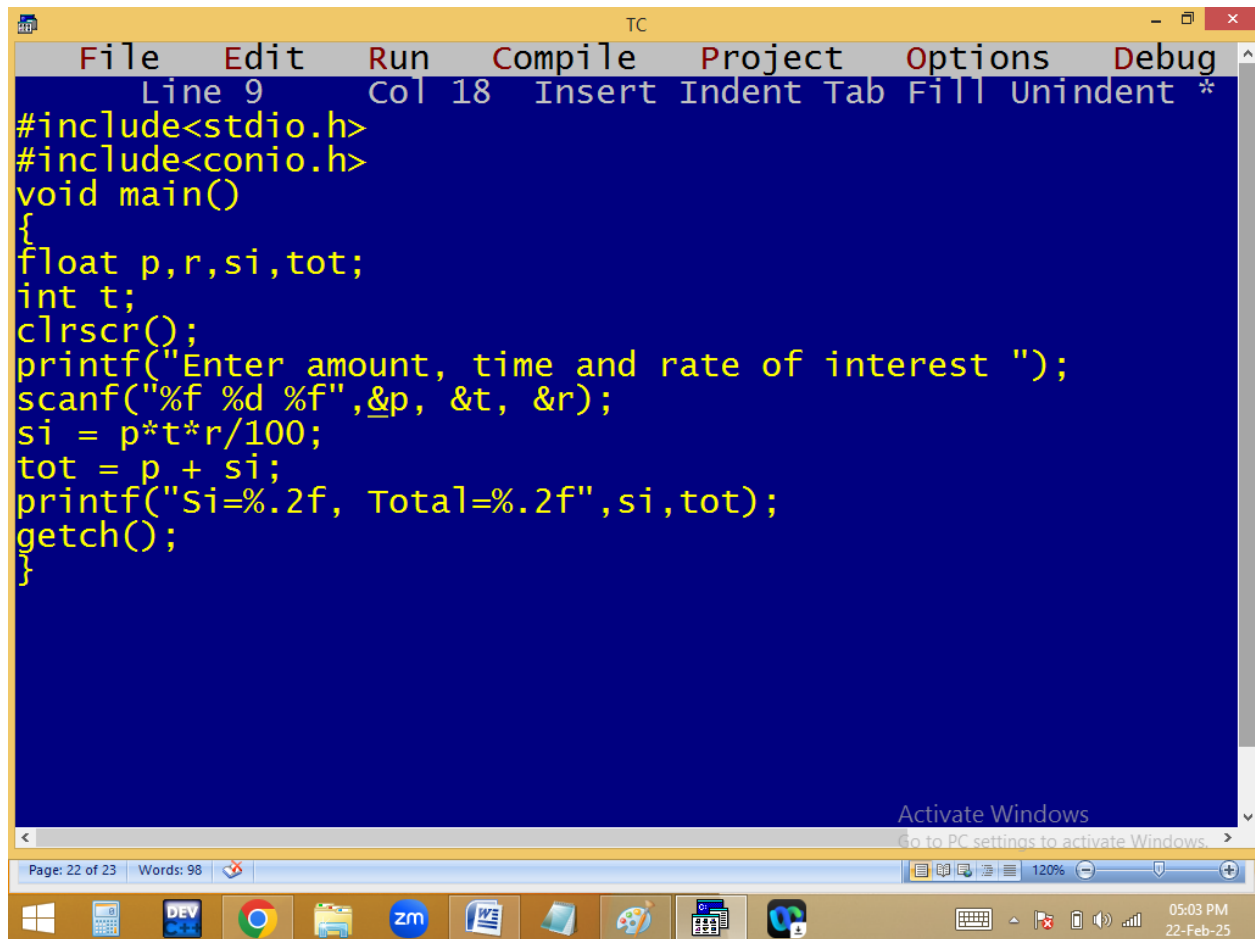
Activate Windows
Go to PC settings to activate Windows.

Page: 21 of 21 Words: 95 120%

04:55 PM
22-Feb-25

Finding simple interest.

$$\begin{array}{ccc} P & R & T \\ 1000 & 2.5 & 10 \end{array} = \frac{SI}{1000}$$
$$10 \times 2.5 = 25 \times 10$$
$$\frac{250}{1000} = 0.25$$
$$1000 + 250 = 1250 - \text{Tot}$$



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 9", "Col 18", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
float p,r,si,tot;
int t;
clrscr();
printf("Enter amount, time and rate of interest ");
scanf("%f %d %f",&p, &t, &r);
si = p*t*r/100;
tot = p + si;
printf("Si=%.2f, Total=%.2f",si,tot);
getch();
}
```

At the bottom of the IDE window, there is a status bar with "Page: 22 of 23" and "Words: 98". Below the IDE window is the Windows taskbar, which includes icons for Windows, File Explorer, DEV, Google Chrome, File Explorer, Zoom, Word, and other applications. The system tray on the right shows the date and time: "05:03 PM 22-Feb-25".

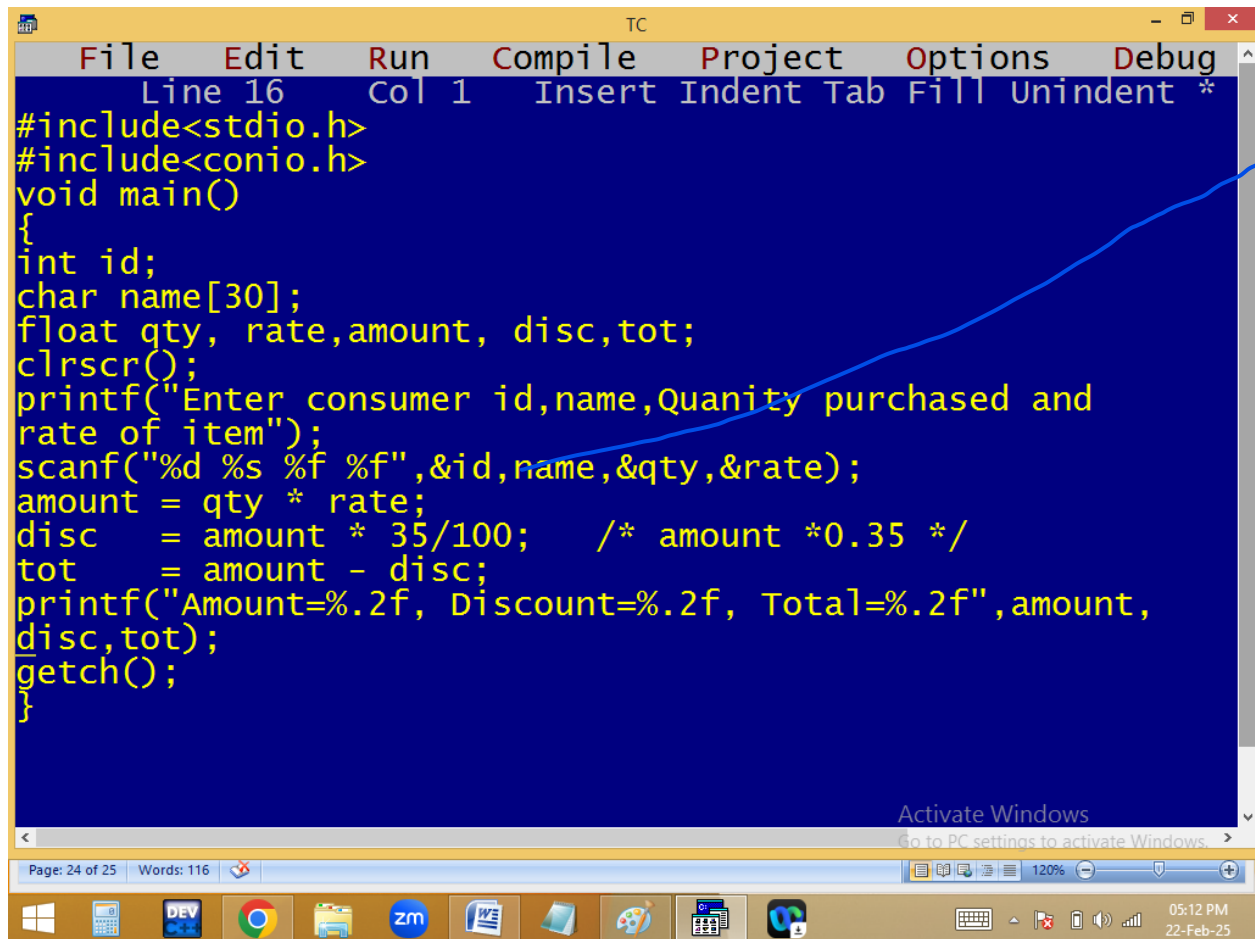
```
TC
Enter amount, time and rate of interest 1000 10 2.5
Si=250.00, Total=1250.00
```

Activate Windows
Go to PC settings to activate Windows.

Page: 23 of 23 Words: 98 120%

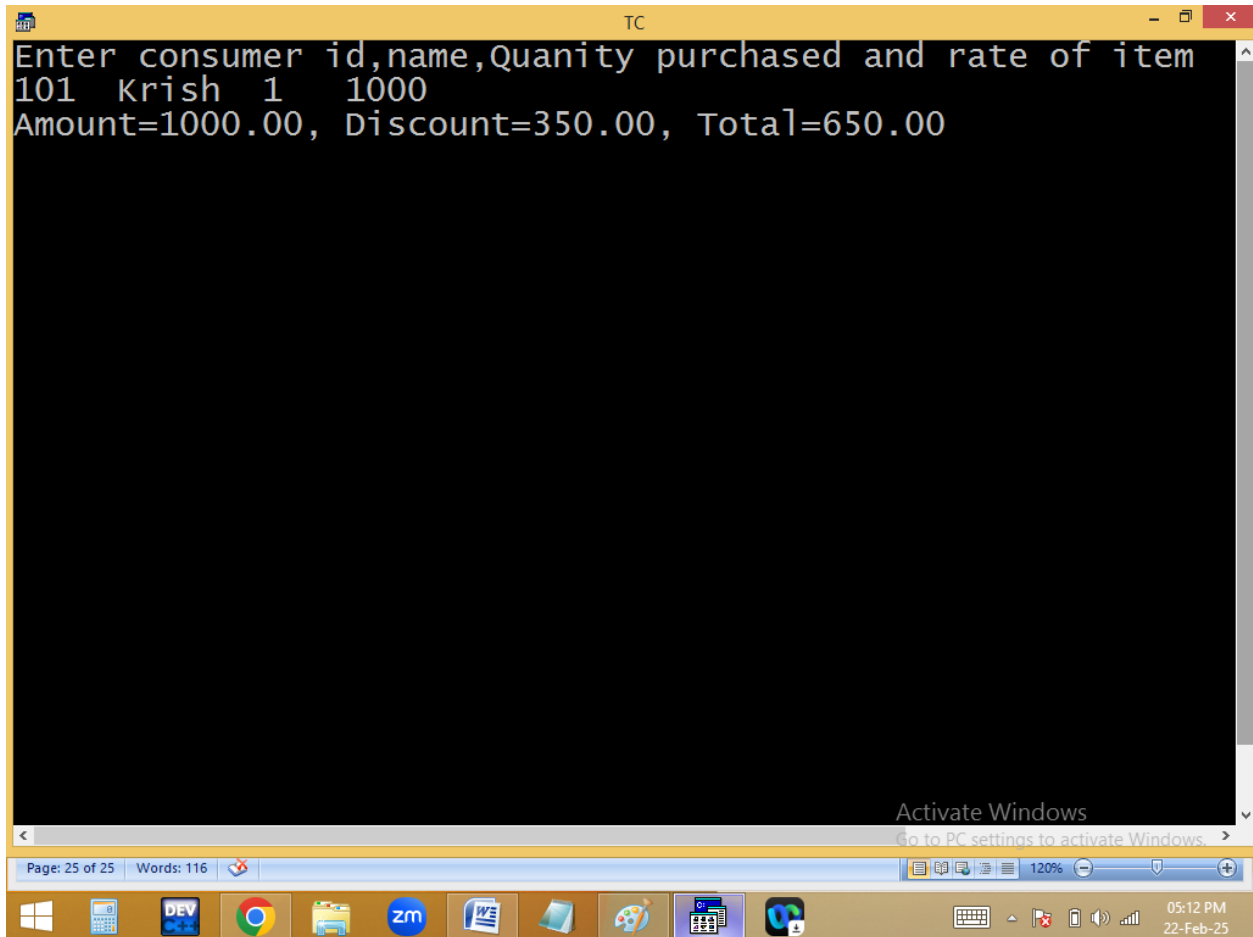
05:03 PM
22-Feb-25

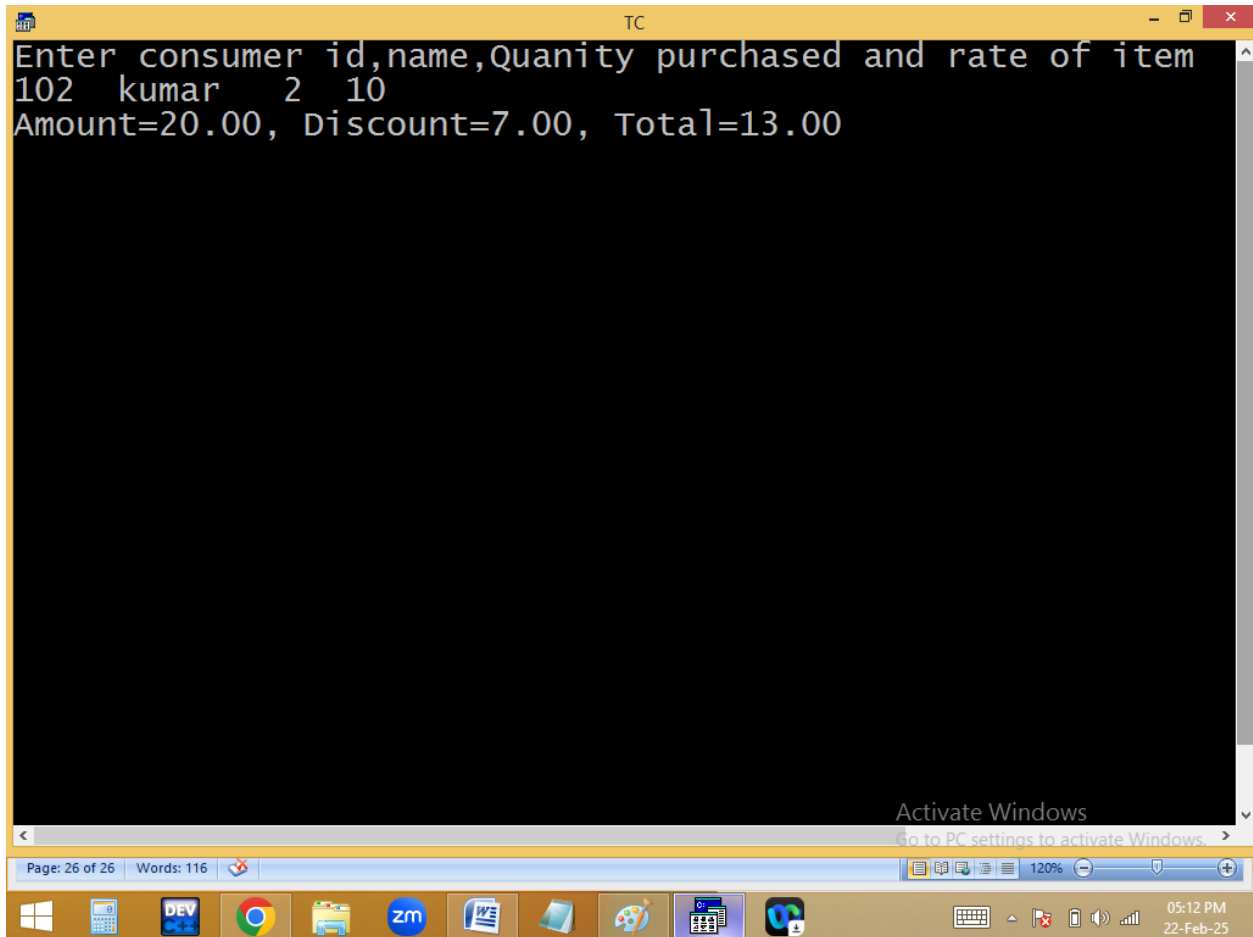
Read a customer id, name, Quantity purchased and rate of item. Find the amount, 35% discount and total.



```
TC
File Edit Run Compile Project Options Debug
Line 16 Col 1 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
int id;
char name[30];
float qty, rate, amount, disc, tot;
clrscr();
printf("Enter consumer id,name,Quantity purchased and
rate of item");
scanf("%d %s %f %f",&id,name,&qty,&rate);
amount = qty * rate;
disc = amount * 35/100; /* amount *0.35 */
tot = amount - disc;
printf("Amount=%.2f, Discount=%.2f, Total=%.2f",amount,
disc,tot);
getch();
}
```

Page: 24 of 25 Words: 116 120% 05:12 PM 22-Feb-25





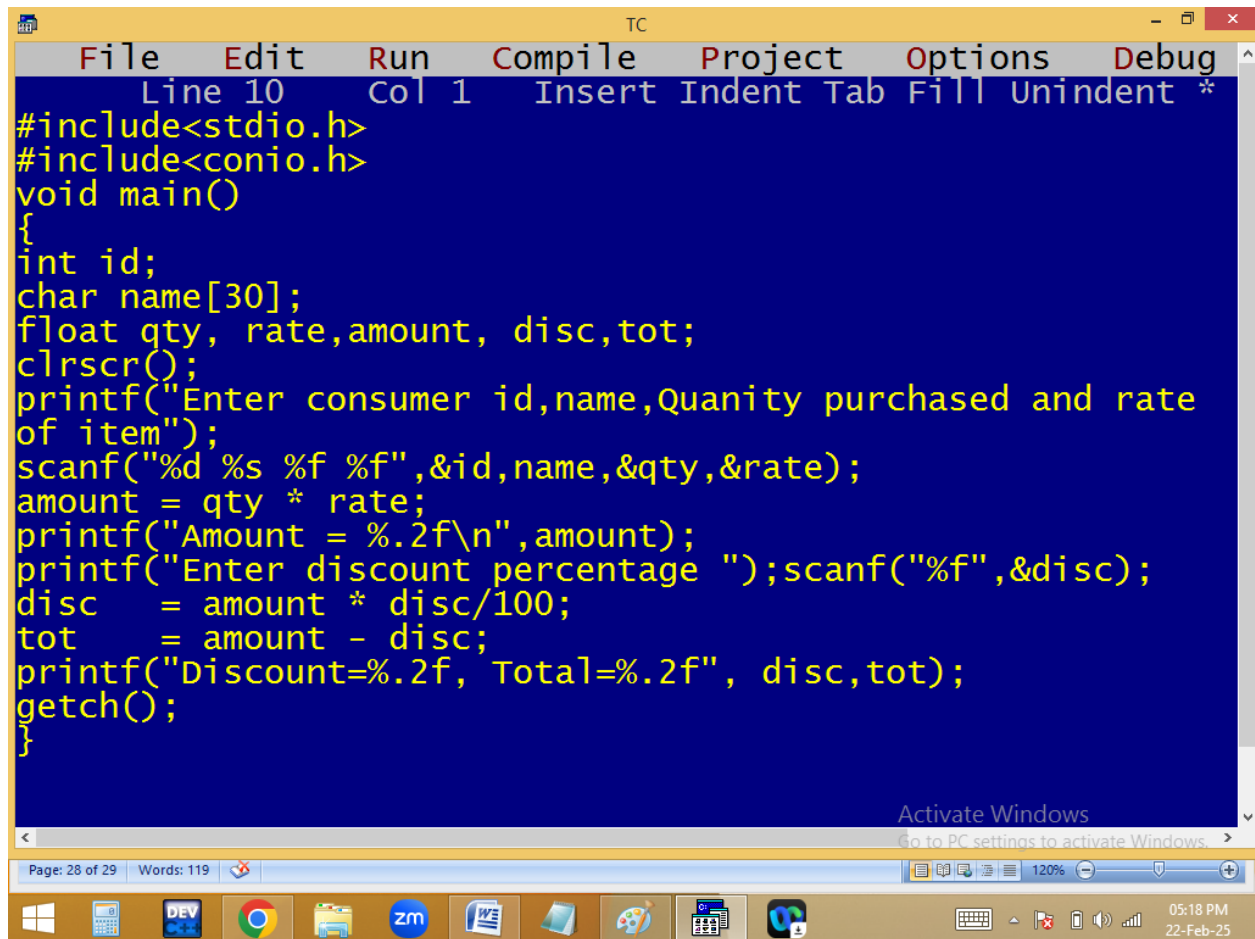
```
TC
Enter consumer id,name,Quantity purchased and rate of item
333 kumari 1 1
Amount=1.00, Discount=0.35, Total=0.65_
```

Activate Windows
Go to PC settings to activate Windows.

Page: 27 of 27 Words: 116 120%

05:14 PM
22-Feb-25

Dynamic [runtime] discount:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 10", "Col 1", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and an asterisk. The main editing area has a dark blue background with yellow text. It contains a C program that calculates the total amount for an item after a discount. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int id;
    char name[30];
    float qty, rate, amount, disc, tot;
    clrscr();
    printf("Enter consumer id,name,Quantity purchased and rate
    of item");
    scanf("%d %s %f %f",&id,name,&qty,&rate);
    amount = qty * rate;
    printf("Amount = %.2f\n",amount);
    printf("Enter discount percentage ");scanf("%f",&disc);
    disc    = amount * disc/100;
    tot     = amount - disc;
    printf("Discount=%.2f, Total=%.2f", disc,tot);
    getch();
}
```

At the bottom of the IDE window, there is a status bar showing "Page: 28 of 29" and "Words: 119". To the right of this, there is a "Go to PC settings to activate Windows" message. The Windows taskbar is visible at the very bottom, showing various application icons and the system clock indicating 05:18 PM on 22-Feb-25.

```
TC
Enter consumer id,name,Quantity purchased and rate of item
1 kumari 1 1
Amount = 1.00
Enter discount percentage 0
Discount=0.00, Total=1.00
```

Activate Windows
Go to PC settings to activate Windows.

Page: 29 of 30 Words: 119 120%

05:18 PM
22-Feb-25

```
TC
Enter consumer id,name,Quantity purchased and rate of item
2 krish 1 100000
Amount = 100000.00
Enter discount percentage 85
Discount=85000.00, Total=15000.00
```

Activate Windows
Go to PC settings to activate Windows.

Page: 30 of 31 Words: 119 120%

05:19 PM
22-Feb-25

```
TC
Enter consumer id,name,Quantity purchased and rate of item
3 wife 1 10000
Amount = 10000.00
Enter discount percentage 100
Discount=10000.00, Total=0.00_
```

Activate Windows
Go to PC settings to activate Windows.

Page: 31 of 32 Words: 119 120%

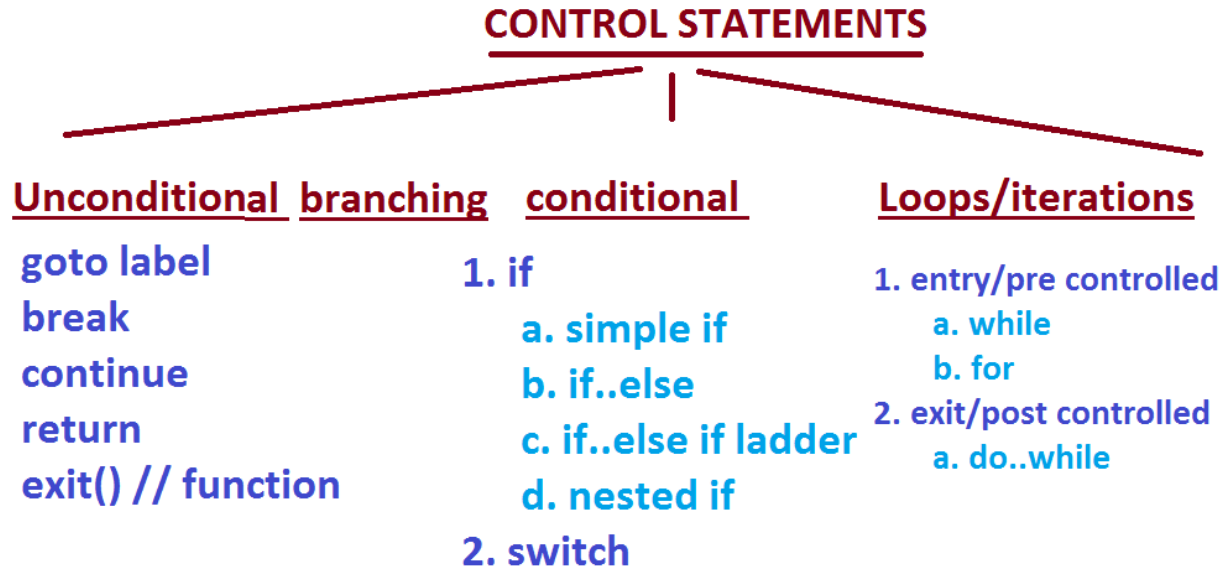
05:19 PM
22-Feb-25


```
TC
Enter consumer id,name,Quantity purchased and rate of item
4 gf 1 1000
Amount = 1000.00
Enter discount percentage 200
Discount=2000.00, Total=-1000.00_

Page: 32 of 33 Words: 119
```

Control statements / control structures

They are used to control program execution flow [order]. In C we are using the following control statements.



goto label / jumping statement

It is used to transfer program execution from one place to another place [label].

In this process it is jumping from one area to another without any condition. Hence it is also called **unconditional** jumping statement.

Syntax:



```
.....;  
.....;  
goto label;  
.....;  
.....;  
label:  
.....;  
.....;
```

The diagram illustrates the execution of a goto statement. A green arrow points down from the first two lines of code to the 'goto label;' statement. Another green arrow points from the 'goto label;' statement to the 'label:' line. A final green arrow points down from the last two lines of code, indicating the flow continues after the label.

Here **goto** is a keyword.

Label is an identifier is used to identify the area[line].

Every label should be end with **:** (colon)

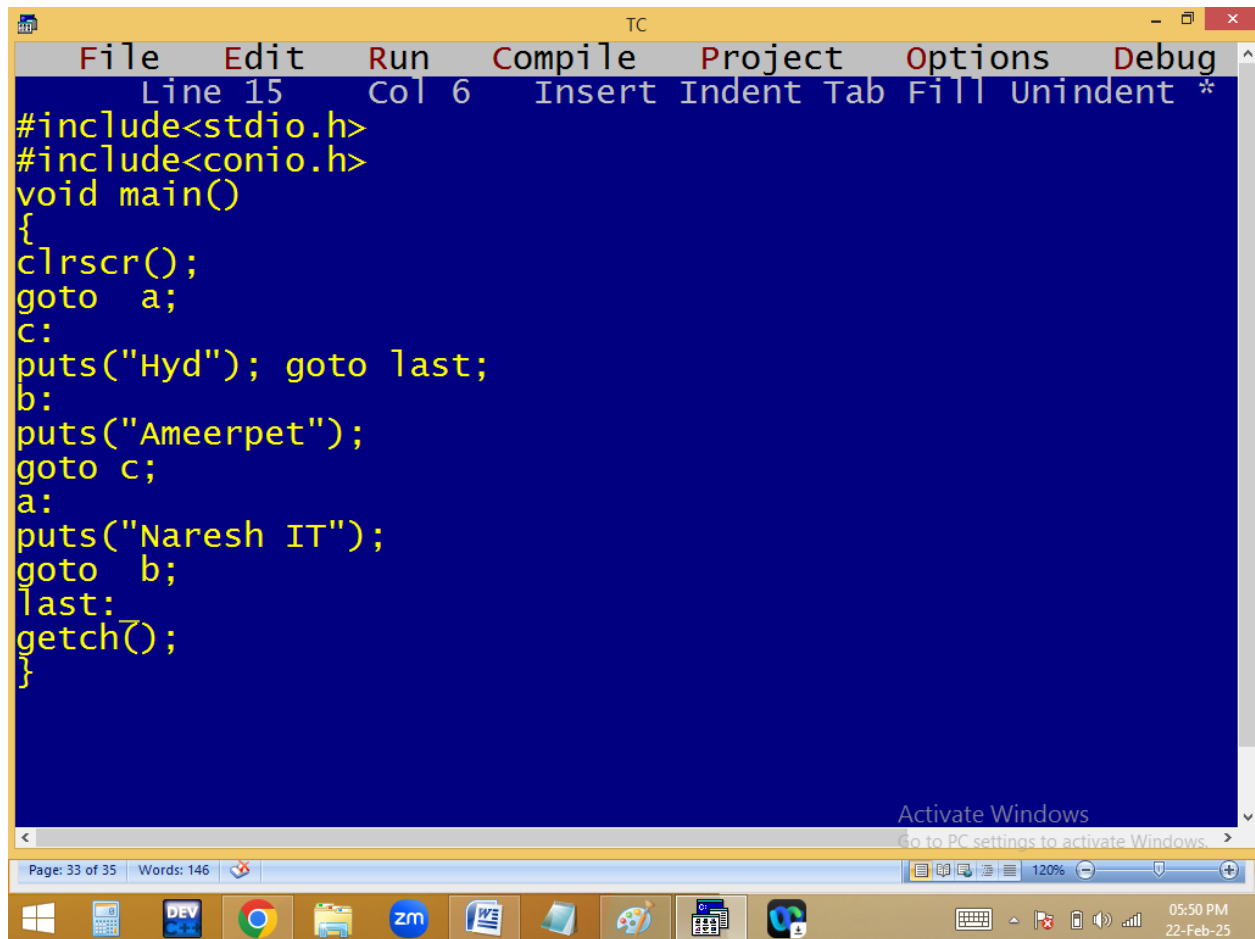
Keywords not allowed in labels i.e. label should be user defined.

Duplicate labels not allowed.

There is no space between go and to.

Label naming rules are similar to the identifier rules.

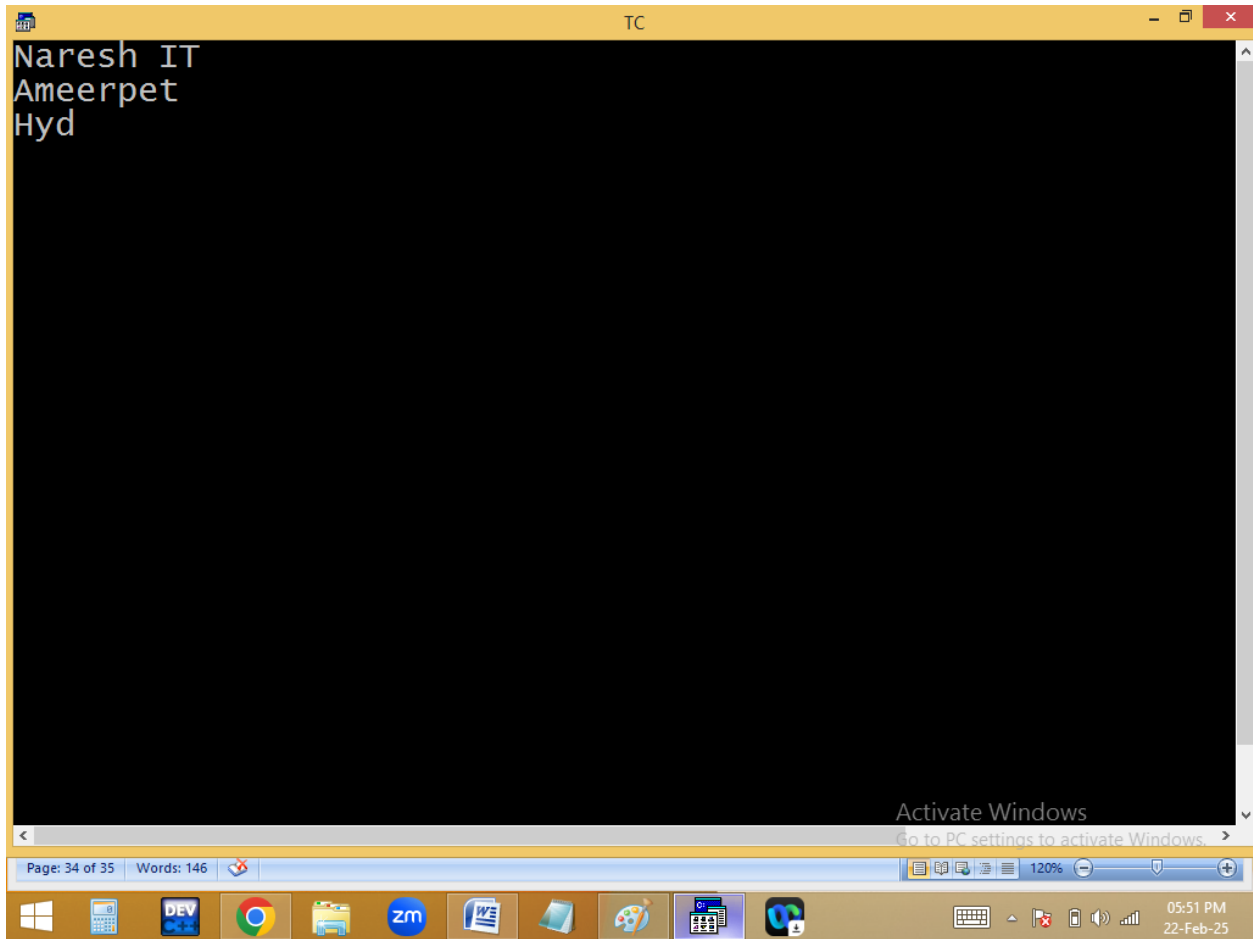
Note: goto label working style is similar to loops some times.

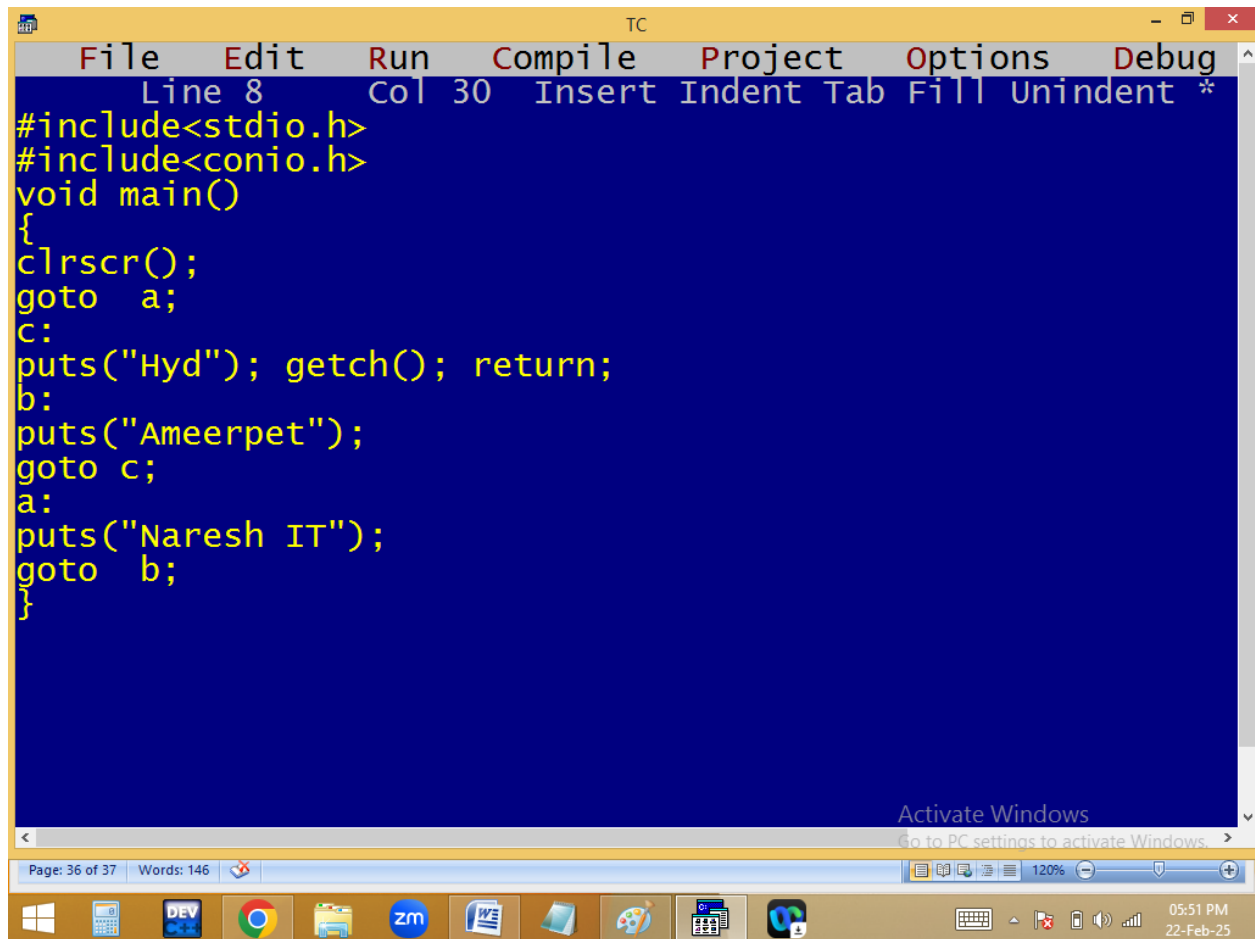


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 15", "Col 6", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
goto a;
c:
puts("Hyd"); goto last;
b:
puts("Ameerpet");
goto c;
a:
puts("Naresh IT");
goto b;
last:
getch();
}
```

At the bottom right of the editing area, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." Below the code editor, a status bar shows "Page: 33 of 35" and "Words: 146". The Windows taskbar is visible at the bottom, showing icons for Windows, Calculator, DEV C++, Google Chrome, File Explorer, Zoom, Word, and other applications. The system clock in the bottom right corner shows "05:50 PM" and "22-Feb-25".

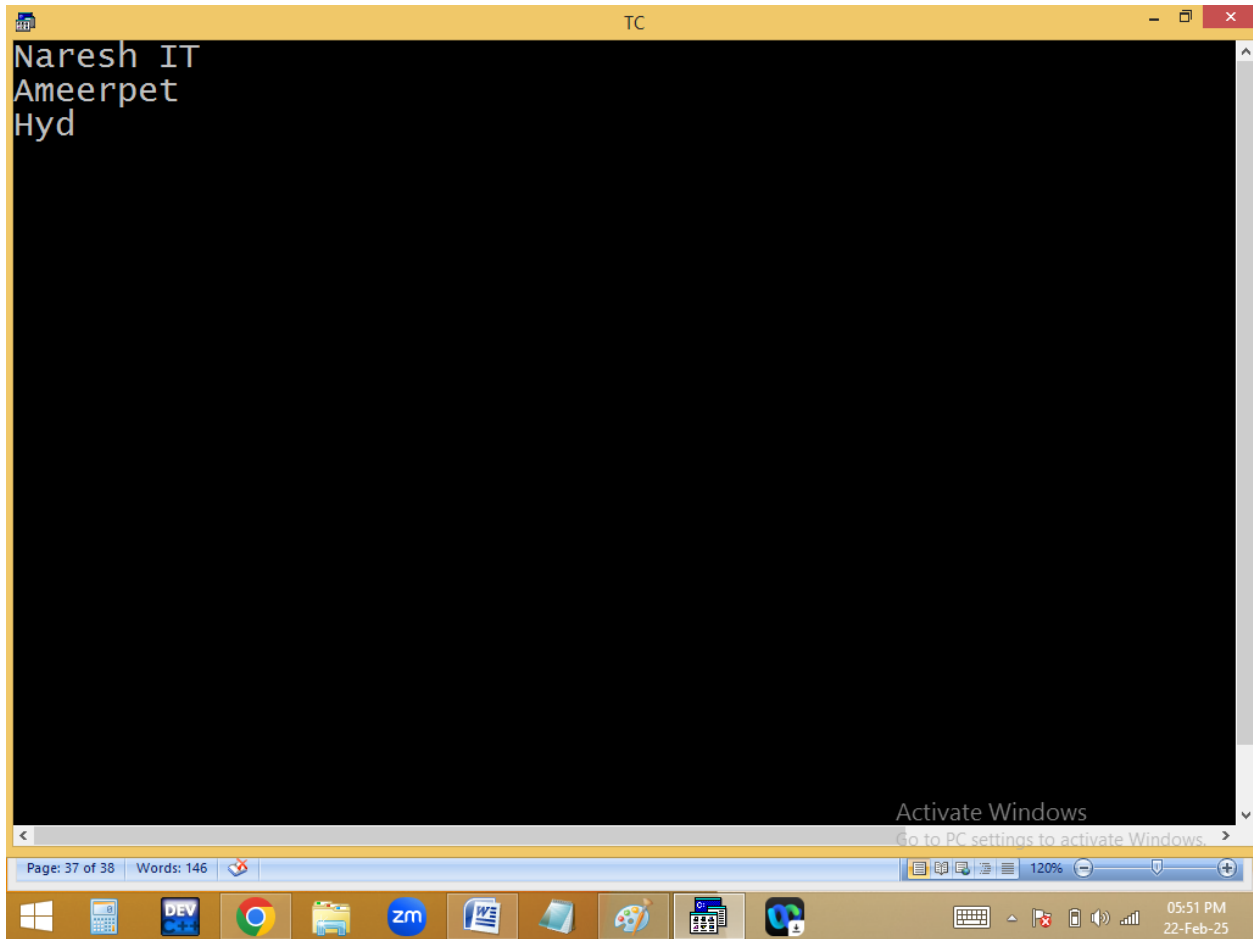


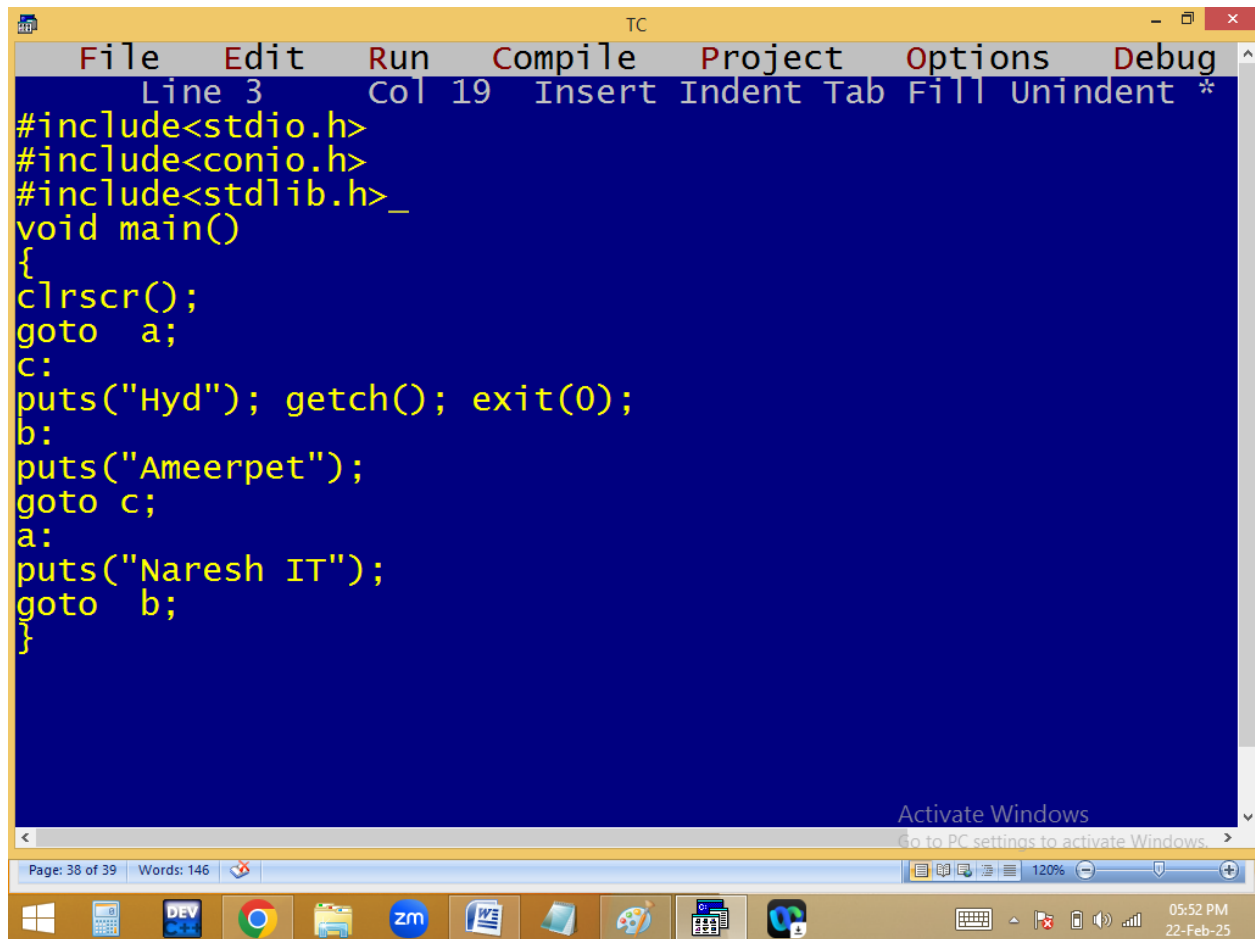


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 8", "Col 30", and lists editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main editing area has a dark blue background with yellow text. It contains a C program that uses goto statements to print a sequence of strings. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
goto a;
c:
puts("Hyd"); getch(); return;
b:
puts("Ameerpet");
goto c;
a:
puts("Naresh IT");
goto b;
}
```

At the bottom of the IDE window, there is a status bar showing "Page: 36 of 37" and "Words: 146". Below the IDE window is the Windows taskbar, which includes icons for the Start menu, Task View, File Explorer, Google Chrome, Zoom, Word, and other applications. The system tray on the right shows the date and time as "05:51 PM 22-Feb-25".





```
TC
File Edit Run Compile Project Options Debug
Line 3 Col 19 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>_
void main()
{
clrscr();
goto a;
c:
puts("Hyd"); getch(); exit(0);
b:
puts("Ameerpet");
goto c;
a:
puts("Naresh IT");
goto b;
}
```

Page: 38 of 39 Words: 146 120%

05:52 PM 22-Feb-25

